

# Multilateral Institutions and Technology Fragmentation: Lessons from the Era of Global Integration

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## Abstract

As multilateral agreements on trade and investment are being reshaped by the changing global economic and political landscape, we examine their historical role in fostering technology interdependence across countries. Using a bilateral technology interdependence measure constructed from semantic relationships between patents over 1980–2020, we document the evolution of countries’ technology around key events, including the collapse of the Soviet Union and the rise of China. We then use an event-study design to estimate the effects of preferential trade agreements (PTAs), bilateral investment treaties (BITs), and WTO accession on bilateral technology interdependence. We find that deep PTAs and BITs lead to a large and statistically significant increase in partner countries’ technology similarity; WTO/GATT accession leads to smaller but still meaningful effects; on the other hand, shallow PTAs have only a modest and mostly noisy effect. These multilateral agreements together account for about 7% of the integration in technology space over 1980 to 2020. Using these estimates, we simulate the technology landscape under counterfactual future multilateral agreements.

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**Keywords:** Technology fragmentation; preferential trade agreements; bilateral investment treaties; WTO accession; patent semantics.

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# 1 Introduction

The global economic and political landscape is increasingly shaped by great-power rivalry, industrial policy, tariff wars, and coercive economic instruments, such as trade and financial sanctions, embargoes, and export controls. The order maintained through multilateral institutions built in the era of integration—preferential trade agreements (PTA), the WTO, and bilateral investment treaties (BIT)—is being challenged and fundamentally reshaped. Motivated by these shifts, a growing literature measures the economic consequences of fragmentation through trade, investment, and capital flows (Aiyar et al., 2023; Gopinath et al., 2024; Alfaro and Chor, 2023; Freund et al., 2024; Campos et al., 2023; Fernández-Villaverde et al., 2024; Cai et al., 2025; Levchenko et al., 2025). The consequences of the shifts, however, could extend beyond traditional outcomes like trade and FDI. In particular, by harmonizing standards, opening common markets, and building predictable cross-border policy environments, integration enabled by multilateral institutions may have pushed firms in different countries toward using similar technologies and standardized and interoperable components. This integration, which takes place in the space of technologies that firms use, allows production linkages across countries to form easily and enables innovators in one country to build on advances made in others. It also confers comparative advantages to countries at the center of the technology space. In cellular wireless, for example, the United States collects billions of dollars in licensing rents on smartphones sold worldwide because of Qualcomm’s dominant share of the standard-essential patents underlying 3G, 4G, and 5G. As multilateral institutions are being weakened, the potential undoing of this technological integration could disrupt production linkages, force duplication of R&D across borders, and ultimately lower aggregate efficiency.

In this paper, we examine the impacts of multilateral institutions on technological interdependence. To speak to the potential impacts of the ongoing disruptions on the technological fragmentation across countries, we draw a lesson from the past, using data from the era of broad global integration. We build a panel of bilateral technological interdependence over 1980–2020 from the semantics of worldwide patents and then ask whether PTAs, BITs, and WTO accession raised the technological integration of country pairs. To the extent they did matter, how much can they account for the technological integration observed in these decades?

To construct this measure, we calculate the distance between semantic meanings of patents obtained as large language model embeddings. We view the embedding of a patent, represented as a vector in a space of 768 dimensions, as capturing the position of the patent in the space of human knowledge. Patents that are close to each other in this space use similar underlying scientific and engineering principles, and are hence more likely to

be interdependent in the sense that firms using these patents could build on each other's products as components or share common input components. As an example, patents by EV companies are much closer to those by battery producers and chemical engineering firms than to those by ICE producers; conversely, patents by ICE producers are closer to those in mechanical engineering. This measure builds on the data we constructed in [Chen et al. \(2025\)](#), where we validate the resulting embeddings against patent-class, input-output, and trade-based measures of technological distance. Here, we extend the measure to cover a larger set of countries and four decades between 1980 and 2020. Moreover, for heuristic interpretation, we conduct a further validation exercise to show that this measure captures genuine relationships in the compatibility of technologies. We discuss how this measure compares with alternative measures, such as similarity in the distribution of patents over patent classes, similarity in frequency of words, and citation intensity.

Building on this measure, we first conduct descriptive case studies of two major events of the past 40 years. The first case is the collapse of the Soviet Union. Former Warsaw Pact members and Soviet republics gradually reduced their economic dependence on Russia, with some, such as Hungary, Poland, and Bulgaria, joining the EU. How did the technological position of these countries evolve? Using our measure, we find that after 1989, Hungary, Poland, and Bulgaria moved sharply toward Western Europe, while Belarus and Moldova stayed close to Russia; Ukraine moved modestly westward only after 2011. This finding mirrors the political alignment observed over these years and suggests that technology is shaped endogenously with the economic and political landscape.

The second case study focuses on the rise of China. China's share of the world economy has risen steadily over the period. We examine the evolution of its technological position over time and compare it with those of other major countries. We find that China has been more distant from the U.S. in the technology space than Germany and Japan, but similar to Russia. China's similarity to the U.S. rises through the 1990s and 2000s, peaks around 2010, and declines after 2012; Russia follows the same pattern, while Germany and Japan remain close to the U.S. throughout.

Since the 2008 financial crisis, the notion of the U.S. and China as the 'Group of Two' has gained traction. We investigate how major countries moved in the space of technology relative to these two countries over this period. Japan and especially Korea have shifted toward the U.S., while Western European economies (Germany, France, and the United Kingdom) stay close to the U.S. Russia stays close to China throughout. These findings suggest that as countries fall into geopolitical blocs, their technologies follow suit.

We isolate specific aspects of global integration and identify their causal effects on countries' technological positions. We focus on three categories: PTAs, BITs, and the WTO. By using an event-study design, we disentangle the role of these institutions from time-

invariant country linkages and other changes in the relationship between pairs of countries. We find that shallow PTAs, agreements that do little more than lower tariffs, have only a modest and noisy effect on bilateral technology. By contrast, deep agreements, which often harmonize standards, simplify regulations, and lower technical barriers to trade, lead to a large and persistent increase in bilateral technology similarity. The effect operates on two margins. First, countries in a PTA tend to specialize in related industries, so their innovation is directed to related patent classes. Second, and perhaps more subtly, the deep integration brought about by PTAs leads countries to use more closely related technologies even within a patent class. To isolate this within-class channel, we introduce two sets of controls. The first control is the cosine similarity of the two countries' patent-class shares. It absorbs any effect that operates purely through countries shifting into similar technological fields. The second is the similarity of their import and export compositions across sectors. It captures their specialization in the space of goods, ruling out that technology similarity merely reflects similarity in what they produce and trade. The deep-agreement effect survives both. We also examine the effect by industry. The deep-PTA effect is larger where standards and technical barriers matter more, with the largest responses in chemicals, pharmaceuticals, computers and electronics, and machinery repair and installation.

We conduct a parallel analysis for BITs and the WTO. BITs may promote technological integration through two mechanisms: by granting access to foreign firms, which bring their own innovation and technology expertise and thereby push countries closer in the technology space; and by harmonizing technical standards and promoting regulatory exchange.<sup>1</sup> We find that, like deep PTAs, BITs lead to a statistically significant increase in bilateral technology similarity, and the effect is robust to the same set of controls (for patent-class composition and trade specialization) described above. The industry-level effects concentrate in the same shared core of regulation-intensive sectors as the deep-PTA effect. For the WTO/GATT, country pairs become more technologically similar after joint accession, but the effects are smaller than for BITs and deep PTAs. The effect also started a couple of years prior to the official accession, consistent with countries undertaking broad market-opening and regulatory reforms during the accession process.

That the effects concentrate where standards and technical barriers matter most points to standards harmonization as a core mechanism. We examine this directly, using data on technology standards and the organizations that set them. We find that partners bound by

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<sup>1</sup>For example, Article 11(8) of the 2012 U.S. Model BIT requires host states to allow foreign investors to participate in the development of product and technical standards by central government bodies. The 2018 Brazil–Ethiopia Cooperation and Facilitation Investment Agreement establishes a Joint Committee and National Focal Points charged with technical dialogue, regulatory exchange, and coordination on technical and environmental regulations. The SADC Protocol on Finance and Investment explicitly aims to “foster harmonisation of the financial and investment policies of the State Parties.”

deep agreements share more similar standards portfolios, and organizations whose members are more integrated issue more standards. As in the technology results, the response is strongest for deep PTAs and BITs and weak for shallow PTAs and the WTO.

We build on the estimates to conduct two sets of counterfactual exercises. Looking back, the integration-era agreements account for 7.0% of the rise in average bilateral technology similarity between 1981–1985 and 2016–2020, with sharper reorganizations in our case studies of Eastern Europe and the rise of China. Looking forward, a China-led realignment in which China deepens its CPTPP and RCEP linkages while the United States exits the multilateral system would push the average highlighted economy by +3.3% of its distance toward the Chinese pole; a mirror US-led realignment via the CPTPP and a revived TTIP, with China excluded from the multilateral system, moves the same economies −3.7% in the opposite direction. To the extent that technology integration supports the integration of global value chains, facilitates the adoption of new technologies, and encourages innovation by expanding markets, our results suggest that the fragmentation of the multilateral order could affect welfare in ways that extend beyond its direct effect on trade and FDI.

Our paper speaks to three strands of work. First, a growing literature constructs measures of geopolitical fragmentation from trade, capital flows, and voting alignment (Aiyar et al., 2023; Gopinath et al., 2024; Fernández-Villaverde et al., 2024; Campos et al., 2023; Cai et al., 2025; Levchenko et al., 2025), and quantifies its consequences for trade flows, supply chains, FDI, and aggregate welfare (Alfaro and Chor, 2023; Freund et al., 2024; Javorcik et al., 2024; Cerdeiro et al., 2021); Baldwin and Freeman (2022) survey the broader landscape of global-supply-chain risk and resilience. We complement these with a bilateral technology object constructed from patent semantics, and show that institutional integration left a technology-space residue that is not exhausted by reallocation of goods trade: controlling for four directional measures of trade-pattern similarity leaves the deep-PTA effect positive and persistent.

Second, our measure of technology interdependence builds on earlier patent-based measures of technology. In particular, Jaffe (1986) measures technological proximity between firms as the cosine similarity of their patent-class share vectors, an approach later generalized by Bloom et al. (2013); Jaffe et al. (1993) trace technological linkages through patent citations; and Kelly et al. (2021) measure innovation from patent textual novelty relative to prior patents and impact on subsequent patents.<sup>2</sup> We capture bilateral *positions* in technology space from the semantics of worldwide patents. Unlike patent-class shares, our measure varies continuously within a patent class; unlike citation-based measures,

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<sup>2</sup>Building on these measures, recent work studies global knowledge diffusion: cross-border research collaboration within multinational firms (Kerr and Kerr, 2018; Bircan et al., 2024) and trade-induced knowledge spillovers (Aghion et al., 2024).

it is symmetric and does not require patents to cite each other; unlike word-frequency measures, it reflects semantic meaning rather than vocabulary only. Using this measure, we document that bilateral technology positions tracked the major economic and political realignments of the past four decades, providing background for recent studies on technology rivalry and geopolitics (Bai et al., 2025; Ma et al., 2025).

Lastly, our investigation into the specific role of multilateral agreements connects to earlier works quantifying the causal effects of such agreements. On PTAs, Baier and Bergstrand (2007) estimate that FTAs raise bilateral trade between members; Mattoo et al. (2022) and Dür et al. (2014) show that deeper agreements generate larger trade effects; Laget et al. (2020) show that deep agreements promote global value chain participation; and Hofmann et al. (2019) document wide variation in PTA depth and content. On BITs, the literature documents that BITs raise FDI and multinational activity (Egger and Merlo, 2012; Hartmann and Spruk, 2023). On the WTO, Subramanian and Wei (2007) document the trade effects of GATT/WTO membership. We add that deep PTAs and BITs both compress bilateral technology distance, with effects concentrated in regulation-intensive sectors; after WTO accession, country pairs likewise become more technologically similar, in a pattern consistent with the broad package of pre-accession reforms.

## 2 Measurement

Before describing our measure, it is helpful to pause and think conceptually about what an ideal measure *should* capture. To speak to countries' integration or fragmentation in the space of technology, the measure should reflect the degree to which the technologies that two countries develop can be combined or extended: whether firms in one country can use the other's outputs as components, whether their products can share input components, and whether innovators can build on each other's advances.

Such interdependence can operate both within and across broad technological fields. Within a field, technologies adhering to a common standard must interoperate, while technologies built on divergent standards do not. For example, every 3G, 4G, and 5G handset implements a pool of standard-essential patents contributed by firms such as Qualcomm, Ericsson, Huawei, Nokia, and Samsung, whose technologies have co-evolved under a common specification. In contrast, Japan's Personal Digital Cellular (PDC) system, standardized by RCR/ARIB and launched commercially by NTT DoCoMo in 1993, was deployed exclusively in Japan through the 1990s and was incompatible with GSM and CDMA-family systems. Across fields, firms with more closely related technologies draw more heavily from each other, while those with unrelated technologies interact less. For example, Tesla's patents sit close to Panasonic's: Tesla's electric vehicles use lithium-ion

batteries from Panasonic, whose cathode-chemistry innovations are important for the energy density that the vehicles require. In contrast, Tesla’s patents sit far from those of diesel engine makers like Cummins and Caterpillar, whose core combustion and fuel-injection portfolios have limited overlap with battery-powered powertrains. These examples highlight the ‘integration/fragmentation of technology’ concept we seek to capture: interoperability and relatedness of countries’ technological portfolios.<sup>3</sup>

## 2.1 Construction from Patent Text

We construct the measure from patent texts. Patents are natural units for measuring technology: each patent documents a specific technological advance, describes its technical antecedents, and states the problem it is designed to solve. We process each patent document through a large-language-model embedding to obtain a dense vector that we interpret as the position of the patent in the space of human technologies.

There are three broad earlier approaches to measuring patent-based technological proximity. The first, following Jaffe (1986) and Bloom et al. (2013), represents each entity by its share vector across patent classes and computes cosine similarity between these vectors. Scalable and transparent, this proximity nonetheless is silent on the within-class component of integration. Our measure, on the other hand, captures both similarity across patent classes and variations after controlling for class composition similarities. The second, the citation-based tradition of Jaffe et al. (1993), traces directed knowledge flows through who cites whom. Citations are asymmetric by construction, and are potentially distorted by strategic non-citation and examiner-added references. They can also miss compatibility altogether. As an example, Tesla’s vehicle thermal-management architecture and Panasonic’s positive-electrode chemistry for lithium-ion cells are inseparable inside an EV, yet again neither cites the other.<sup>4</sup> Our measure is instead symmetric and picks up the closeness of these patents in the space of technology. The third tradition, exemplified by Kelly et al. (2021), measures textual content from document–term matrices. This captures lexical overlap but not semantic content: it misses synonyms, paraphrasing, and negation, so close in vocabulary need not mean close in meaning. Our measure based on semantic embeddings

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<sup>3</sup>It is a horizontal measure of the distance between countries in technology space, as opposed to a vertical measure of their total innovative capacity. Two countries with different portfolio sizes can be close in technology space if their portfolios point in the same direction.

<sup>4</sup>In a lithium-ion EV, cathode chemistry is a primary determinant of the cell’s safe operating window. At the upper end, layered cathodes (NMC, NCA) decompose and release oxygen, accelerating thermal runaway; at the lower end, lithium plating on the graphite anode during charging accelerates degradation. The vehicle thermal-management architecture exists to hold cell temperature inside that window across ambient and load conditions, so cathode chemistry and thermal-management design are co-engineered components of the same product. Tesla’s thermal-management patent: US10967702B2. Panasonic’s cathode-chemistry patent: US10833317B2.



of full-sentence patent text alleviates these concerns.

This measure extends the construction in [Chen et al. \(2025\)](#) to the universe of world patents published between 1980 and 2020 and to a broader country sample. We next describe the patents, the formal definition of the measure, and the aggregation from the patent level to country and country-sector units.

The patents are drawn from the EPO PATSTAT Global database, 2023 Autumn edition, which assembles bibliographic and full-text records from the major patent offices worldwide; we embed all patent documents published between 1980 and 2020 using the Alibaba gte-base-en-v1.5 sentence-encoder model (see Appendix A for details). The cosine similarity between any two patent embeddings is a pairwise measure of technological proximity. For each country  $d$  and year  $t$  we take the mean of the embeddings of patents originating in  $d$  in year  $t$  and normalize the resulting vector to unit length; we denote the country-year embedding by  $\hat{v}_{dt}$ . The bilateral technology similarity between countries  $d$  and  $o$  in year  $t$  is the cosine similarity between their country-year embeddings,  $S_{dot} = \hat{v}_{dt} \cdot \hat{v}_{ot}$ , and the corresponding bilateral distance is defined as

$$D_{dot} = \sqrt{2(1 - S_{dot})} = \|\hat{v}_{dt} - \hat{v}_{ot}\|, \quad (1)$$

the Euclidean distance between the two vectors.<sup>5</sup> The same construction extends to finer and coarser units: for the country-sector-year heterogeneity specifications we restrict the average to patents classified in sector  $j$ ; for the regional case studies in Section 3, where we aggregate countries to regions such as West Europe, we extend the average to patents originating in any country in the region.

## 2.2 A Heuristic Example: Vehicle Powertrains

We use the vehicle sector as an example to illustrate how our measure captures the position of patents, and by extension of firms and countries, in the space of human technology. Firms in this industry occupy different parts of that space, and the suppliers they draw on sit in correspondingly different positions. We show that the embedding-based measure captures these positions as intended.

We focus specifically on powertrains. Battery-electric and internal-combustion-engine (ICE) are two powertrains supported by different technologies. They also rely on different upstream technologies: batteries on one side, fuel-supply systems on the other. Our measure, if working as intended, should be able to achieve two goals. First, within the class

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<sup>5</sup>Bootstrap resampling at the patent level shows that  $D_{dot}$ —the distance between the mean patent embeddings of  $d$  and  $o$ —is close to the average pairwise distance between individual patent embeddings across the two countries,  $\frac{1}{N_{dt}N_{ot}} \sum_{i \in d, j \in o} \|e_i - e_j\|$ , while being orders of magnitude cheaper to compute.



of powertrain patents, the measure should separate the patents that correspond to ICE from those that correspond to EV. Second, across patent classes, the measure should show higher closeness between the EV powertrain patents and battery patents, and between the ICE powertrain patents and fuel-system patents.

To implement the test, we need to map patents to technologies in an intuitive way. We work at the IPC subclass level (IPC4). Within each IPC4 we partition the patents into a small number of *technology clusters* using *K*-means on the patent embeddings.<sup>6</sup> A cluster is a set of patents whose embeddings are close to each other within a given IPC4, so it isolates a coherent technological theme finer than the IPC4 itself. We label each cluster by its top Term Frequency–Inverse Document Frequency (TF-IDF) keywords—words that appear often in the cluster but rarely in sibling clusters within the same IPC4.<sup>7</sup> We refer to each cluster by its top keywords.

We consider patents within three IPC4s related to the vehicle production chain. *B60K* (Arrangement or Mounting of Propulsion Units or Drive Control in Vehicles), henceforth “Vehicle Powertrain”, contains patents on how propulsion is organized and controlled inside a vehicle and therefore pools both EV and ICE powertrain technologies. *H01M* (Processes or Means for the Direct Conversion of Chemical Energy into Electrical Energy, e.g., Batteries), henceforth “Battery”, contains battery-cell technologies. *F02M* (Supplying Combustion Engines with Combustible Mixtures or Constituents Thereof), henceforth “Fuel Supply”, contains fuel-system technologies for combustion engines. *B60K* sits downstream of both *H01M* and *F02M*: a vehicle’s powertrain draws on batteries when it is electric and on fuel-supply systems when it is combustion-based.

Figure 1 plots the technology clusters inside each of the three IPC4s. Each IPC4 is drawn as a circle; clusters are dots labeled by their top TF-IDF keywords. We highlight clusters whose keywords overlap with those of a cluster in a neighboring IPC4 and label the rest in grey with only the top keyword. Two groups stand out inside *B60K*. The first, in green, contains clusters whose top keywords include *battery box*, *battery module*, *charging control*, and *recharging*, all of which also appear among the top keywords of clusters in *H01M*. The second, in red, contains clusters whose top keywords include *fuel container*, *fuel tank*, *fuel vapor*, and *filler pipe*, with the same vocabulary recurring in *F02M*. The upstream side mirrors the downstream: *H01M* clusters centered on *battery module* and *thermal management* share vocabulary with the *B60K* battery-labeled clusters, and *F02M* clusters centered on *evaporative emission* and *fuel pump* share vocabulary with the *B60K* fuel-labeled clusters. With only patent text as input, the measure separates the EV and ICE groups

<sup>6</sup>For each IPC4 with at least 20 patents we set the number of clusters to scale with the size of the class, capped between 5 and 30, and fit *K*-means on the embeddings. The precise rule is reported in Appendix A.

<sup>7</sup>Intuitively, TF-IDF ranks a word highly if it is characteristic of the cluster, that is, common inside it and uncommon elsewhere within the same IPC4. The exact score is defined in Appendix A.

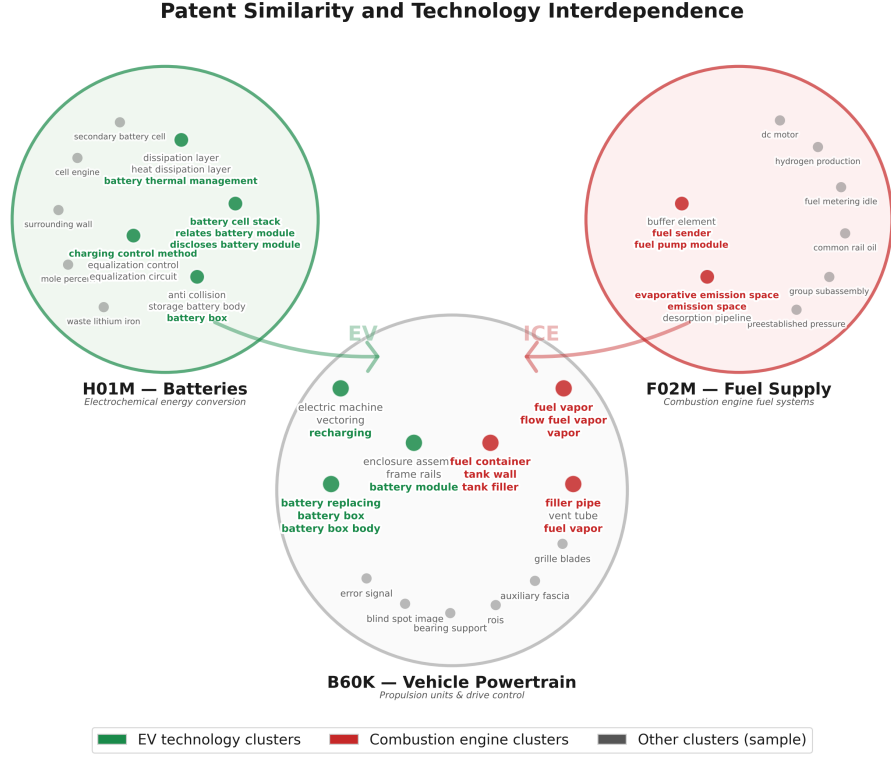


Figure 1: Technology clusters: vehicle powertrain and upstream classes.

*Notes:* Each circle represents one IPC4 class (B60K: Vehicle Powertrain; H01M: Battery; F02M: Fuel Supply). Dots inside each circle are technology clusters obtained by K-means on patent embeddings within the IPC4, labeled by their top TF-IDF keywords relative to sibling clusters in the same IPC4. Clusters whose keywords overlap with clusters in a neighboring IPC4 are highlighted and labeled with their top three keywords; a representative sample of the remaining clusters is shown in grey with only the top keyword.

inside B60K and aligns each group with the upstream class whose technology it draws on.

To measure the distance we apply the similarity measure in equation (1) at the cluster level. Table 1 reports the cosine similarity between each group of B60K clusters and the two upstream IPC4s. The first row averages the embeddings of the three B60K clusters that share keywords with H01M (the EV clusters) and reports their similarity with H01M and with F02M. The second row does the same for the three B60K clusters that share keywords with F02M (the ICE clusters). As a benchmark, the last row reports the similarity between the full B60K class (pooling all clusters) and each of the two upstream IPC4s.

At the IPC4 level, B60K as a whole is more similar to F02M (0.864) than to H01M (0.720), reflecting the fact that combustion vehicles still account for the bulk of vehicle patents in the sample. Once we condition on the technology clusters within B60K, the two upstream classes separate more sharply. The three B60K clusters with battery-related keywords are about equally close to the battery class and the fuel-supply class (0.772 vs. 0.768), closing—

Table 1: Cluster-Level Similarity between B60K and Its Upstream Classes

| B60K unit                                 | Similarity with |                    | # Patents |
|-------------------------------------------|-----------------|--------------------|-----------|
|                                           | H01M (Battery)  | F02M (Fuel Supply) |           |
| Battery-keyword-overlapping clusters (EV) | 0.772           | 0.768              | 2,468     |
| Fuel-keyword-overlapping clusters (ICE)   | 0.679           | 0.893              | 2,258     |
| Full B60K (benchmark)                     | 0.720           | 0.864              | 25,703    |

*Notes:* Each row represents a subset of B60K patents (rows 1 and 2) or all B60K patents (row 3). For each row, we compute the weighted average of the embeddings of patents in the subset, and report its cosine similarity with the weighted-average embedding of all patents in H01M and in F02M, respectively. Weights are the IPC4 shares of each patent. Embeddings are based on patent abstracts filed during 2000–2014.

and marginally reversing—the IPC4-level gap in favor of F02M. The three B60K clusters with fuel-related keywords are 0.214 more similar to F02M than to H01M (0.893 vs. 0.679), amplifying the IPC4-level gap.

This example illustrates the two implications from the opening paragraphs of this section: within a patent class that pools EV and ICE powertrain patents, the measure separates them; across the vertical chain, it aligns each group with the upstream class that supplies it—evidence that the measure captures interoperability, not just lexical overlap.

In [Chen et al. \(2025\)](#), we further validate the measure along three further dimensions: bilateral similarity is higher between patents in the same patent class, between patents whose sectors are closer in input-output space, and between country pairs that trade more intensively with each other.

### 2.3 Summary Statistics

Figure 2 summarizes the cross-sectional distribution of bilateral technology similarity at the country-pair-year level. The measure is concentrated around the middle of the unit interval, with a mean of 0.546 and a standard deviation of 0.199, but it also has substantial mass in the upper tail. Figure 3 plots mean bilateral technology similarity over 1980–2020, both raw and residualized on the IPC4 patent-class cosine similarity; the comparison separates the component of bilateral similarity that co-moves with patent-class composition from the within-class component that the embeddings are designed to recover.

Table 2 reports the numbers of country pairs, country-sector pairs, and patents underlying each unit, alongside summary statistics for the dependent and independent variables of the event-study analysis.

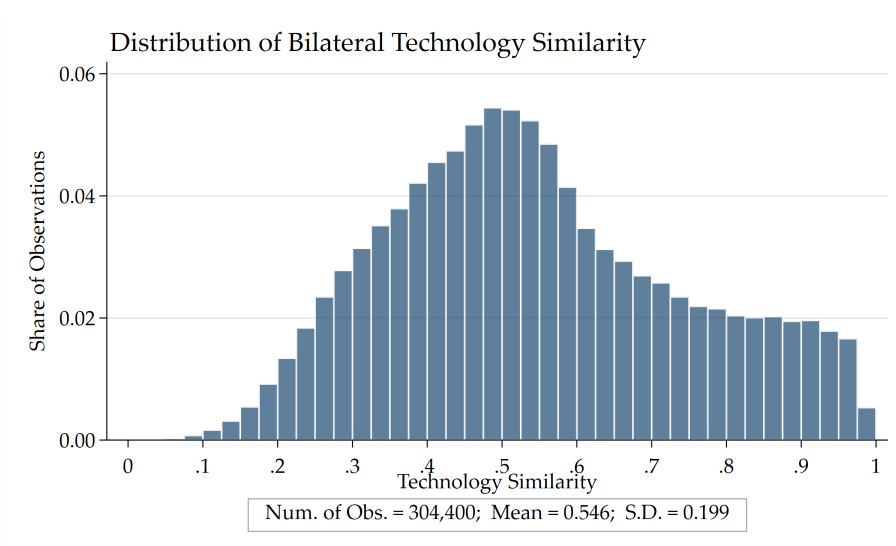


Figure 2: Distribution of bilateral technology similarity

Notes: The vertical axis reports the share of directed country-pair-year observations in each technology-similarity bin.

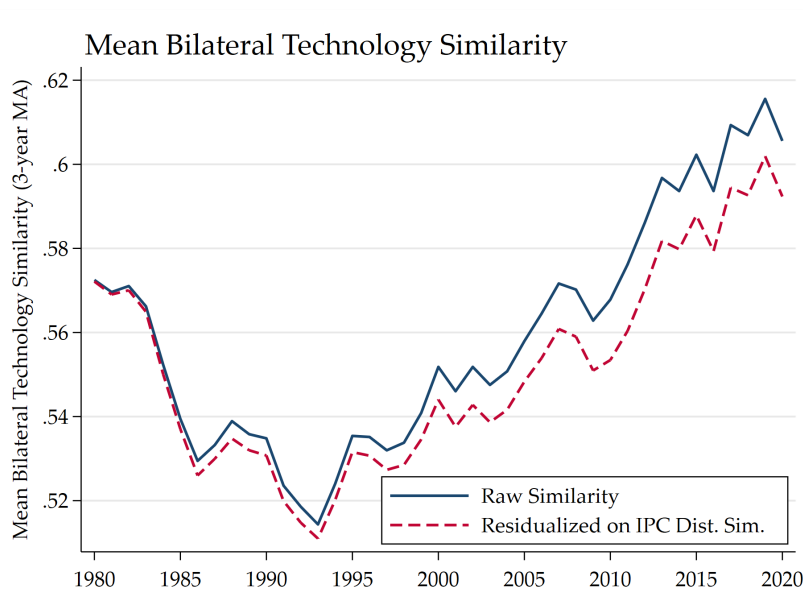


Figure 3: Mean bilateral technology similarity over time

Notes: The figure plots three-year moving averages of raw bilateral technology similarity and similarity residualized on IPC4 patent-class cosine similarity.

### 3 Case Studies: Movement in Technology Space

Building on the measure, we conduct case studies on the evolution of the technological positioning of countries around key economic and geopolitical events. The first episode

Table 2: Sample composition and summary statistics.

| <b>Panel A. Units underlying the measure</b>    |                         |                      |        |                                                  |       |        |       |           |
|-------------------------------------------------|-------------------------|----------------------|--------|--------------------------------------------------|-------|--------|-------|-----------|
|                                                 | N                       | Mean                 | SD     | Min                                              | p25   | Median | p75   | Max       |
| Years                                           | 41                      |                      |        |                                                  |       |        |       |           |
| Countries                                       | 261                     |                      |        |                                                  |       |        |       |           |
| Sectors                                         | 83                      |                      |        |                                                  |       |        |       |           |
| Country Pairs                                   | 28,319                  |                      |        |                                                  |       |        |       |           |
| Country Pair-years                              | 304,400                 |                      |        |                                                  |       |        |       |           |
| Country Pair-sector-years                       | 4,897,240               |                      |        |                                                  |       |        |       |           |
| Patents/country-year                            | 4,863                   | 3,798                | 42,330 | 0.05                                             | 2.0   | 12.7   | 158   | 1,464,773 |
| <b>Panel B. Outcome and continuous controls</b> |                         |                      |        |                                                  |       |        |       |           |
|                                                 | N                       | Mean                 | SD     | Min                                              | p25   | Median | p75   | Max       |
| Tech. sim., $S_{dot}$                           | 304,400                 | 0.546                | 0.199  | 0.005                                            | 0.400 | 0.523  | 0.683 | 1.000     |
| Tech. sim., sector, $S_{dojt}$                  | 4,897,240               | 0.556                | 0.179  | -0.025                                           | 0.426 | 0.540  | 0.678 | 1.000     |
| IPC4 cossim, IPC4CosSim $_{dot}$                | 276,312                 | 0.149                | 0.180  | 0.000                                            | 0.005 | 0.081  | 0.227 | 1.000     |
| Trade cossim, exp-exp                           | 160,830                 | 0.130                | 0.192  | 0.000                                            | 0.010 | 0.047  | 0.162 | 1.000     |
| Trade cossim, imp-imp                           | 160,100                 | 0.435                | 0.240  | 0.000                                            | 0.232 | 0.417  | 0.630 | 0.993     |
| Trade cossim, exp-imp                           | 160,468                 | 0.226                | 0.229  | 0.000                                            | 0.042 | 0.145  | 0.345 | 0.994     |
| Trade cossim, imp-exp                           | 160,464                 | 0.227                | 0.227  | 0.000                                            | 0.043 | 0.146  | 0.355 | 0.995     |
| <b>Panel C. Agreement coverage</b>              |                         |                      |        |                                                  |       |        |       |           |
|                                                 | # pairs<br>ever-treated | Share of<br>pair-yrs | Mean   | <i>Within-pair share of years (ever-treated)</i> |       |        |       |           |
|                                                 |                         |                      |        | Min                                              | p25   | Median | p75   | Max       |
| Shallow PTA                                     | 5,373                   | 0.264                | 0.793  | 0.063                                            | 0.621 | 0.964  | 1.000 | 1.000     |
| Deep PTA                                        | 604                     | 0.035                | 0.662  | 0.122                                            | 0.567 | 0.634  | 0.854 | 1.000     |
| BIT                                             | 2,229                   | 0.129                | 0.699  | 0.025                                            | 0.561 | 0.714  | 0.875 | 1.000     |
| Joint GATT/WTO                                  | 12,202                  | 0.545                | 0.861  | 0.083                                            | 0.759 | 1.000  | 1.000 | 1.000     |

Notes: Country-pair and country-sector-pair statistics, 1980–2020. Sectors are ISIC Rev. 4 2-digit. Patent counts are fractional, splitting each patent across inventor countries, IPC4 classes, and (for the sector level) ISIC sectors. Trade cosine similarities are available from 1995. Panel C: shallow / deep PTA = DESTA depth  $\geq 1$  /  $\geq 4$ ; columns 3–6 describe the within-pair share of in-sample years covered among ever-treated pairs.

we study concerns the changes in former Warsaw Pact members and Soviet republics after the collapse of the Soviet Union. The second episode is the integration of China into the world economy since 1990.

### 3.1 Eastern Europe After 1989

After the collapse of the Soviet Union, former Warsaw Pact members and Soviet republics gradually integrated with Western Europe economically. We ask whether their positions in the space of technology followed a similar reorientation, and we measure each country's position between the Western European and Russian poles.

For country  $d$  and year  $t$ ,  $\bar{S}_{d,WE,t}$  is the weighted average of  $d$ 's bilateral similarities to the Western European economies, with weights proportional to each Western European

economy's 1980 patent count;  $\bar{S}_{d,RUS,t}$  is  $d$ 's bilateral similarity to Russia.<sup>8</sup> Each of these similarities converts to a bilateral distance via equation (1). To evaluate where each of these economies sits relative to the two poles, we project its position in technology space onto the line connecting Western Europe and Russia, and normalize that line to unit length so that Western Europe sits at one endpoint and Russia at the other. Formally, the projection for country  $d$  in year  $t$  is given, by the triangle identity, by

$$x_{dt} = \frac{D_{d,WE,t}^2 + D_{RUS,WE,t}^2 - D_{d,RUS,t}^2}{2 D_{RUS,WE,t}^2} = \frac{1 - \bar{S}_{d,WE,t} + \bar{S}_{d,RUS,t} - \bar{S}_{RUS,WE,t}}{2 (1 - \bar{S}_{RUS,WE,t})}. \quad (2)$$

A value of  $x_{dt} = 0$  places  $d$  at the Western European pole and  $x_{dt} = 1$  at the Russian pole; intermediate values locate  $d$  between the two. The scalar thus summarizes each country's position on the Western-Europe–Russia line in a single number per year.

Figure 4 plots  $x_{dt}$  for selected Eastern European economies over 1981–2020. Hungary, Poland, and Bulgaria move from intermediate positions in the early 1990s toward the Western European pole over the subsequent two decades, alongside their institutional integration with Western Europe: Europe Agreements entering into force in the mid-1990s, EU accession for Hungary and Poland in 2004 and for Bulgaria in 2007. Belarus and Moldova, which remained outside these arrangements, stay relatively close to the Russian pole throughout. Ukraine shifts modestly toward the Western pole after 2011, coinciding with the 2014 Euromaidan revolution and Ukraine's EU Association Agreement signed the same year.

The reorientation in  $x_{dt}$  reflects both reallocation of patenting across broad technological fields and convergence within a given field. To isolate the within-field component, we residualize each bilateral similarity on the country pair's IPC4 patent-class cosine similarity and recompute the projection; the reorientation survives residualization, so the within-field component accounts for a non-trivial share of the movement.

### 3.2 The Rise of China

China's integration into the world economy accelerated after Deng Xiaoping's 1992 Southern Tour, which reaffirmed the reform-and-opening policy, and continued through a sequence of multilateral commitments, most notably WTO accession in 2001. We first examine how China's technologies evolved in relation to those of the United States over this period, and compare China to selected benchmark economies.

Figure 5 plots the average technology similarity of China, Germany, Japan, and Russia

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<sup>8</sup>Pre-1991 Russia coding: USSR-era patents are assigned to the present-day inventor country via the PAT-STAT inventor-address records. Sensitivity to alternative weighting schemes for the WE pole (e.g., equal weights or 2000 patent counts) is small in unreported checks.



Figure 4: Technology position of Eastern European economies, 1981–2020.

Notes: The horizontal axis is the calendar year. The vertical axis is  $x_{dt}$  from equation (2); smaller values indicate closer alignment with Western Europe, larger values with Russia.

to the United States. China’s similarity to the United States rises steadily through the 1990s and 2000s, peaking around 2010. After 2012, however, China’s similarity declines sharply; Russia follows the same pattern, with its similarity to the United States also falling after 2012. In contrast, Germany and Japan, traditional U.S. allies, remain close to the United States throughout the period, with only modest upward drift.

Since the 2008 financial crisis and China’s continued rise in the world economy, the notion of a ‘Group of Two’ of the United States and China has gained traction in policy discussions. Figure 5 tracks bilateral similarities to the United States, while Figure 6 takes a complementary view: it projects a broader set of economies onto the US–China line using equation (2), with the United States at the  $x = 0$  pole and China at the  $x = 1$  pole, so that each country’s position is summarized by a single scalar that captures how it is placed between the two poles.

Two patterns emerge. Cross-sectionally, Western European economies (Germany, France, and the United Kingdom) sit close to the U.S. pole; Japan occupies the middle of the line; and Korea and Russia sit closer to the Chinese pole. Over time, Japan and Korea move toward the U.S. pole: Korea in particular shifts from a position near China in 2007–2009 to an



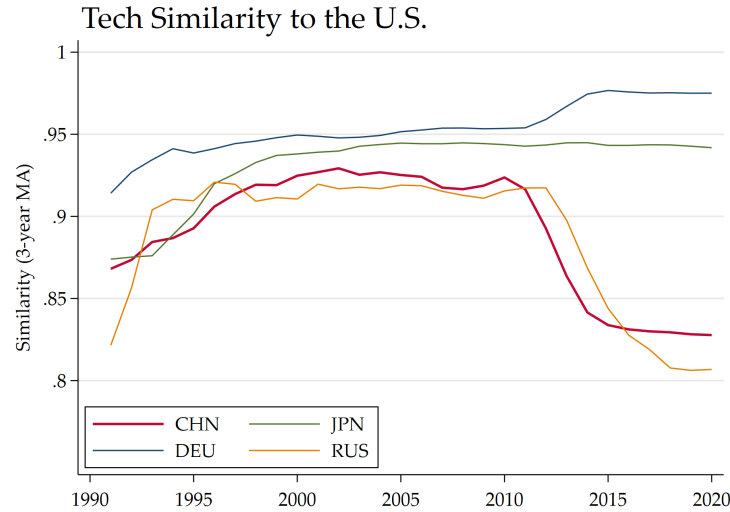


Figure 5: Technology similarity to the United States, 1991–2020.

Notes: Three-year moving average of bilateral similarity to the United States. Data span 1991–2020.

interior position by 2019–2021. Russia stays relatively close to the Chinese pole throughout.

The two case studies show that countries’ positions in technology space move over time, and the movements often track their economic integration and geopolitical alignment. Section 4 examines empirically the role of multilateral agreements (PTAs, BITs, and the WTO) in shaping these positions.

## 4 Multilateral Agreements and Technology Alignment

We investigate the causal effects of multilateral agreements—preferential trade agreements (PTAs), bilateral investment treaties (BITs), and WTO accession—on countries’ positions in technology space. Our strategy is an event-study design around the dates on which country pairs enter each agreement.

We obtain bilateral PTA entry dates and the Economic Integration Agreement (EIA) depth classification from the NSF-Kellogg Institute Data Base on Economic Integration Agreements<sup>9</sup>; BIT entry-into-force dates from the UNCTAD International Investment Agreements Navigator; and WTO and GATT joint-membership dates from the Gravity database maintained by CEPII (Conte et al., 2022). The identification strategy also requires bilateral controls for concurrent changes in trade patterns. We construct these from the BACI international trade database maintained by CEPII (Gaulier and Zignago, 2010), which reports

<sup>9</sup><https://kellogg.nd.edu/nsf-kellogg-institute-data-base-economic-integration-agreements>.



Figure 6: Technology positions on the US–China line, 2007–2021.

Notes: Construction follows equation (2), with the United States and China as the two poles. Smaller values place a country closer to the United States pole; larger values closer to the Chinese pole.

bilateral trade at the 6-digit HS product level, and aggregate to four directional cosine similarities over 6-digit product shares.

Figure 7 shows the expansion of the agreement network over the sample period. Shallow PTA coverage expands steadily, with a visible increase in the early 2000s, rising from 15.1% of country pairs in 1995 to 25.4% in 2015. Deep PTA coverage grows more slowly, increasing from 0.83% in 1995 to 2.13% in 2015, reflecting the narrower reach of agreements with extensive regulatory and institutional provisions. BIT coverage also rises gradually from the 1990s onward, reaching a share of 9.11% of country pairs by the end of the sample. Joint GATT/WTO membership is the broadest institution, expanding sharply during the 1990s and early 2000s and covering around 45% of country pairs by the end of the sample.

Table 2 reports the sample composition and summary statistics. The estimation sample covers 261 countries and 83 sectors over 1980–2020, yielding 304,400 country-pair-years across 28,319 country pairs and 4,897,240 country-pair-sector-years. Panel A details the units underlying the measure; Panel B summarizes the outcome variables and the continuous controls used in the regressions; Panel C summarizes the agreement coverage.

The remainder of the section lays out the regression specification and reports the evi-

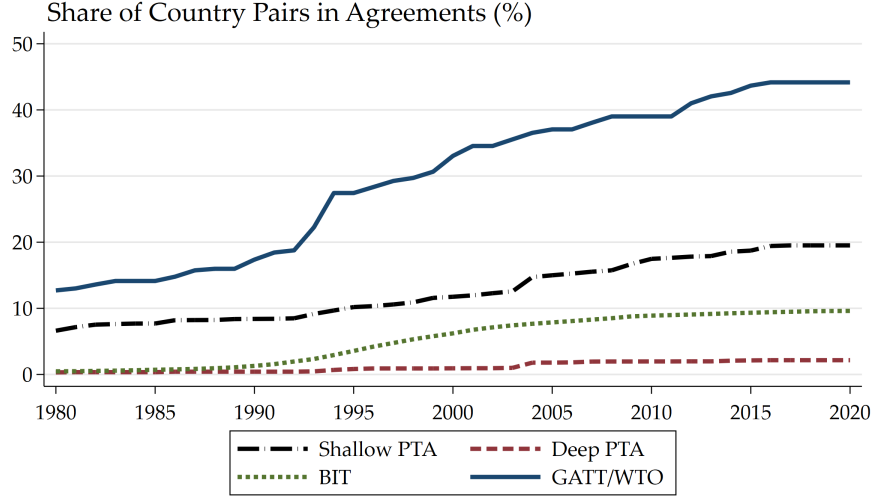


Figure 7: Coverage of multilateral agreements over time.

Notes: The vertical axis reports the share of undirected country pairs covered by each agreement type, in percentage points. Once a pair enters an agreement category, it remains covered in later years.

dence for each agreement.

#### 4.1 Empirical Specification

The bilateral technology similarity  $S_{dot}$  from Section 2 is the outcome in all specifications, and the unit of observation is the country pair  $(d, o)$  in year  $t$ . For each agreement, the running variable is event time  $s \equiv t - T_{do}^I$ , where  $T_{do}^I$  is the first year in which  $d$  and  $o$  are jointly covered by agreement  $I \in \{\text{PTA shallow}, \text{PTA deep}, \text{BIT}, \text{WTO}\}$ . The regression at the country-pair-year level is

$$S_{dot} = \sum_{s \neq -1} \beta_s^I \mathbb{I}[d \text{ and } o \text{ are in } I] \mathbb{I}[t - T_{do}^I = s] + \text{FE}_{do} + \text{FE}_{dt} + \text{FE}_{ot} + \epsilon_{dot}. \quad (3)$$

For shallow and deep PTAs we run (3) as a stacked regression with both treatments. For sector-level exercises we replace the unit  $(d, o, t)$  with  $(d, o, j, t)$  and the fixed effects with  $\text{FE}_{doj} + \text{FE}_{djt} + \text{FE}_{ojt}$ , which absorb any  $(d, o, j)$ -specific level, any  $(d, j, t)$  shock, and any  $(o, j, t)$  shock.

We define *shallow* PTAs as agreements with Baier–Bergstrand Economic Integration Agreement (EIA) depth between 1 and 3 (non-reciprocal preferential arrangements and free trade areas), and *deep* PTAs as agreements with depth  $\geq 4$  (customs unions, common markets, and economic unions). We define the BIT event year as the first year a bilateral investment treaty enters into force between the two countries. WTO/GATT accession is

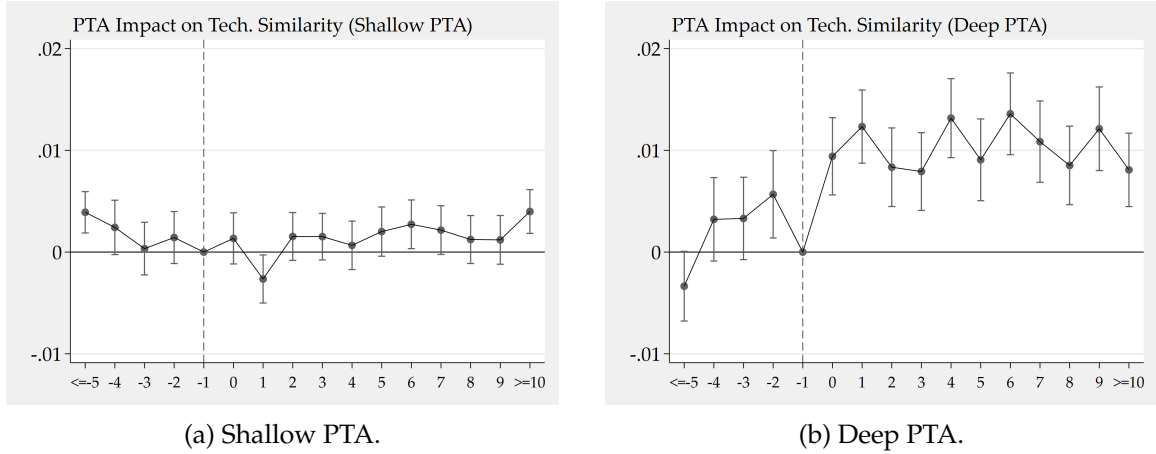


Figure 8: PTA effects on bilateral technology similarity.

*Notes:* Dependent variable is  $S_{dot}$ . Specifications follow equation (3); the shallow panel uses shallow-PTA entry and the deep panel uses deep-PTA entry. Fixed effects:  $do$ ,  $dt$ ,  $ot$ . Bars show 95% confidence intervals based on standard errors clustered at the country-pair level. Horizontal axis is event time relative to agreement entry, with pre-treatment periods at  $s \leq -5$  and post-treatment periods at  $s \geq 10$  grouped.

defined by the first year  $d$  and  $o$  are jointly in the WTO or GATT.

Standard errors are clustered at the country-pair level for country-pair specifications and at the country-pair-sector level for sector-level specifications.

## 4.2 Preferential Trade Agreements

We begin with PTAs, the most common of the three agreement types. PTAs range from shallow tariff-preference arrangements to deep agreements that harmonize technical regulations, extend mutual recognition of standards, and bind partners to regulatory-cooperation disciplines (Hofmann et al., 2019; Laget et al., 2020). We exploit this variation by separating shallow from deep PTAs and asking whether the two differ in their effects on bilateral technology similarity.

**Country-pair event study.** Figure 8 plots the shallow-PTA response and the deep-PTA response separately. The shallow panel estimates (3) around entry into any shallow PTA. The deep panel estimates the response around entry into deep PTAs, i.e., the coefficients are not an incremental effect on top of a shallow-PTA baseline. The pre-trends are mostly flat and statistically indistinguishable from zero. After entry, the shallow response is small and noisy, while the deep-PTA response rises quickly and remains positive for at least a decade. This effect is materially the same when we estimate the event study with a staggered DID design.

**Unpacking the effects.** The response of  $S_{dot}$  to deep PTA entry has two margins. Country pairs can become more similar in the space of technology because they reallocate

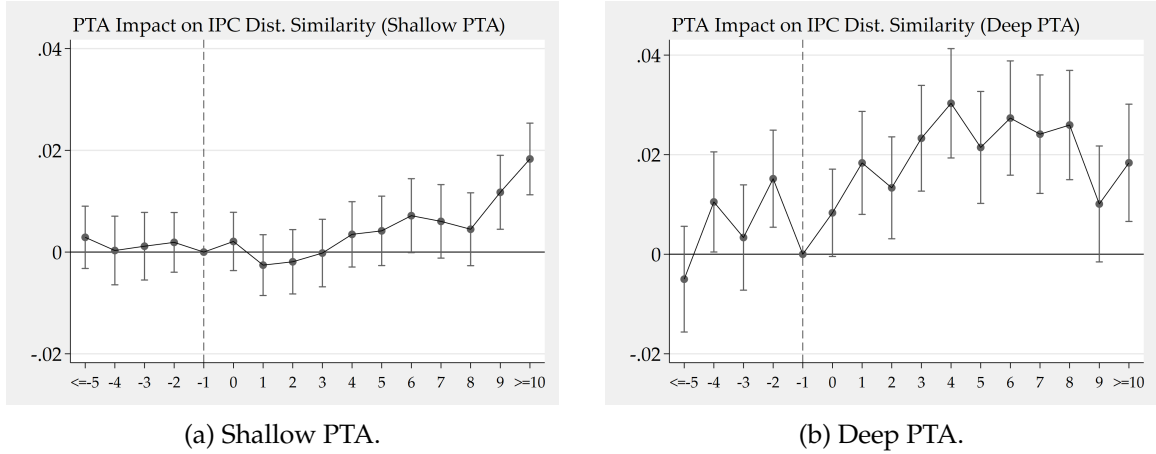


Figure 9: PTA effects on IPC4 patent-share similarity.

Notes: Outcome is  $IPC4CosSim_{dot}$ , the cosine similarity between the IPC4 patent-share distributions of  $d$  and  $o$  (mean 0.148, standard deviation 0.180). Specifications follow equation (3).

patenting across broad technological fields (IPC4 classes), or because their technologies are becoming more similar to each other even conditioning on the IPC distribution. Figures 9 and 10 unpack the two margins in turn.

The first margin, across-field reallocation, could simply be a free-trade effect. Since shallow and deep PTAs both raise bilateral trade between members (Baier and Bergstrand, 2007), which can shift the composition of patenting toward overlapping broad technological fields, they both can have an effect on this margin. Figure 9 reports the event-study response of the IPC4 patent-share cosine similarity  $IPC4CosSim_{dot}$ , the canonical patent-class proximity measure in the tradition of Jaffe (1986) and Bloom et al. (2013).

The left panel plots the shallow-PTA response: IPC similarity rises gradually and persistently after entry. The natural interpretation is the trade-driven composition channel: tariff reductions raise bilateral trade, and bilateral trade pulls each partner's patenting toward the technological fields in which the other specializes.

The right panel plots the deep-PTA response, which is faster and larger. We attribute the larger and more persistent deep-PTA effect to the regulatory-cooperation and standards-harmonization provisions that distinguish deep from shallow agreements (Hofmann et al., 2019; Laget et al., 2020): these disciplines make cross-border production easier than tariff preferences alone, especially for technology-intensive exchanges, further amplifying the across-field reallocation.

Figure 10 looks into the within margin by controlling for IPC composition and bilateral trade patterns in the baseline specification for deep PTA. Specifically, Panel (a) includes  $IPC4CosSim_{dot}$  as a contemporaneous control, absorbing the across-field reallocation of Figure 9; the remaining response is positive, gradual, and persistent, so deep PTAs align

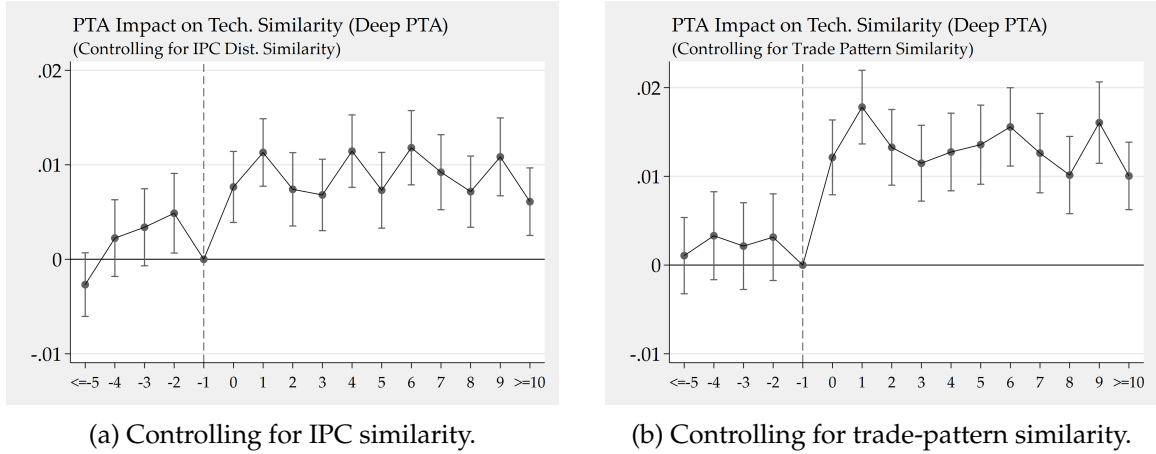


Figure 10: Deep-PTA response on  $S_{dot}$  with IPC and trade-pattern controls.

Notes: Outcome is  $S_{dot}$  in both panels. Panel (a) estimates the deep-PTA event study with  $IPC4CosSim_{dot}$  as a contemporaneous control. Panel (b) estimates the deep-PTA event study with the four directional trade-pattern cosine similarities (export–export, import–import, export–import, import–export) as contemporaneous controls. Specifications otherwise follow equation (3).

technologies within patent classes as well as across them. Panel (b) instead controls for the similarity of bilateral trade patterns, constructed as four directional cosine similarities between  $d$  and  $o$  over 6-digit HS product shares (export–export, import–import, export–import, and import–export): the first two capture convergence in goods baskets, the second two the vertical reorganization of supply chains. The deep-PTA response on  $S_{dot}$  remains positive and persistent under these controls, so the effect cannot be explained away by convergence in the structure of bilateral trade alone.

**Industry-level heterogeneity.** We next replace the country-pair-year panel with a country-pair-sector-year panel.  $j$  indexes 2-digit ISIC (Rev. 4) sectors, and the fixed effects become  $FE_{doj} + FE_{djt} + FE_{ojt}$ . Figure 11 plots post-entry deep-PTA coefficients by 2-digit ISIC sector as a horizontal coefplot with 95% confidence intervals. We use non-manufacturing as the base category, which we exclude from the figure.

We find positive and significant deep-PTA effects in chemicals (ISIC 20), pharmaceuticals (21), non-metallic minerals (23), computers and electronics (26), and machinery repair and installation (33). These are sectors where technical standards, certification, and regulatory coordination play a documented role in cross-border production. Indeed, in Figure 12, we plot the estimated coefficient against the importance of technological compatibility for an industry, constructed as the share of patents in the industry that explicitly mention words like “compatibility” or “interoperability” in their text. We find that industries for which technological compatibility is more important show larger deep-PTA effects on technology interdependence.

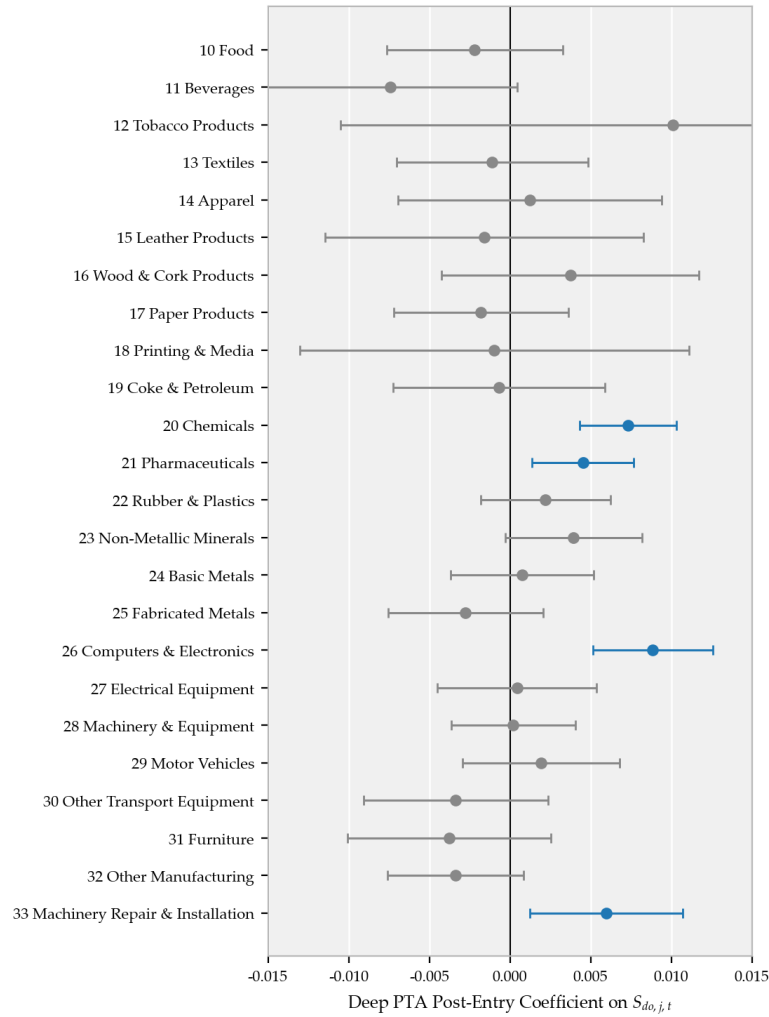


Figure 11: Deep-PTA effect by 2-digit ISIC sector.

*Notes:* Coefficients are post-entry deep-PTA effects by 2-digit ISIC sector, excluding the omitted non-manufacturing base category. Horizontal bars are 95% confidence intervals based on robust standard errors clustered at the country-pair-sector level. Blue markers denote coefficients positive and significant at the 5% level or better.

Our finding supports the notion that, by promoting harmonization in standards and regulations and by making it easier for cross-border production to take place, deep PTAs drive technology integration.

### 4.3 Bilateral Investment Treaties

We next turn to BITs. BITs protect foreign investors and reduce uncertainty around investment conditions. By promoting foreign direct investment, they open a first channel onto technology space: better investment conditions support multinational production



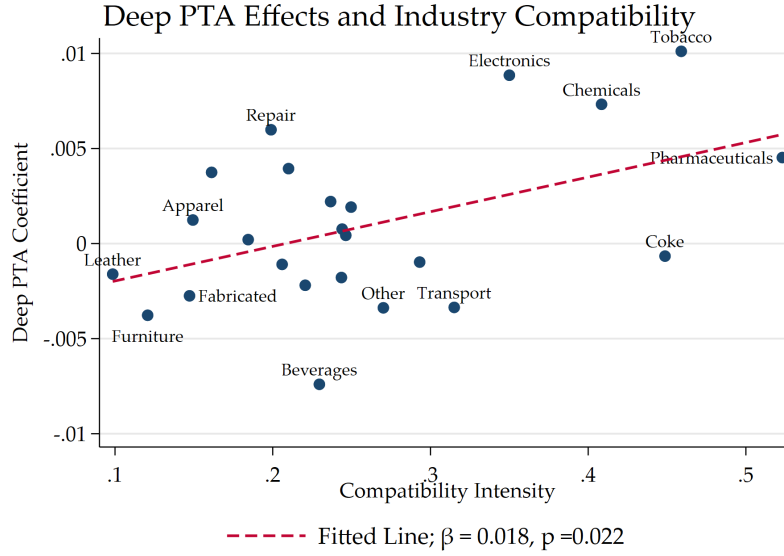


Figure 12: Deep-PTA sector effects and industry compatibility.

Notes: Each marker is a 2-digit ISIC manufacturing sector. The vertical axis plots the sector-specific post-entry deep-PTA coefficient from the country-pair-sector-year DiD specification. The horizontal axis plots the company-based compatibility intensity measure. The dashed line is the fitted linear relationship across sectors.

(Egger and Merlo, 2012; Hartmann and Spruk, 2023), and multinationals transmit practices, supplier requirements, and process know-how across borders, pulling the technological portfolios of host and home countries together. In addition to investor protection, many modern BITs also include transparency, standards-setting participation, and harmonization language; the 2012 U.S. Model BIT, the Brazil–Ethiopia Cooperation and Facilitation Investment Agreement (CFIA), and SADC’s Protocol on Finance and Investment are examples. This leads to a second mechanism: harmonized regulation and standards let firms in partner countries use more compatible technologies, pulling partner countries closer in technology space.

Figure 13 reports the country-pair BIT event study, using specification (3) with a single BIT event-time indicator. The BIT response is positive, gradual, and persistent, with a post-entry peak that is similar to the deep-PTA peak.

Figure 14 reports the BIT event study under the two controls used for deep PTAs: patent-class composition (IPC4 cosine similarity) and trade-pattern similarity. The left panel adds  $IPC4CosSim_{dot}$  as a contemporaneous control; the BIT response is materially the same as in Figure 13. The country-pair response thus survives conditioning on both concurrent PTAs and IPC4 patent-class composition, consistent with investment-mediated integration as a distinct force from the trade channel that drives the shallow-PTA response.

At the industry level, Figure 15 reports the post-entry BIT coefficients by 2-digit ISIC

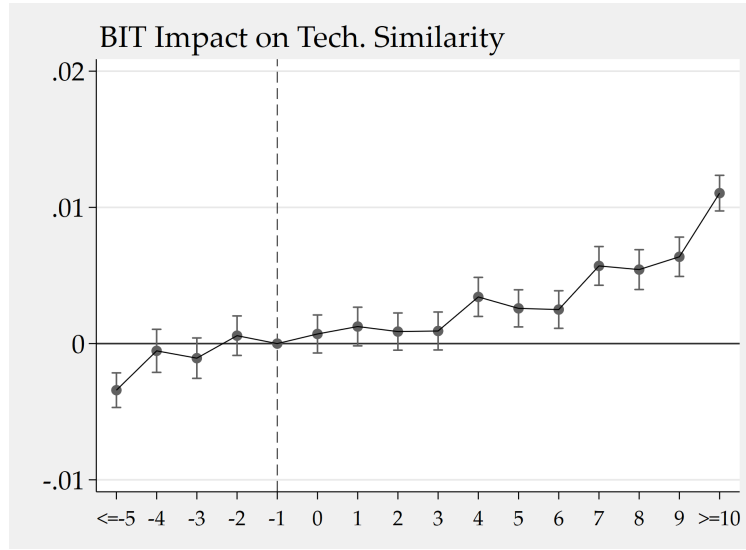
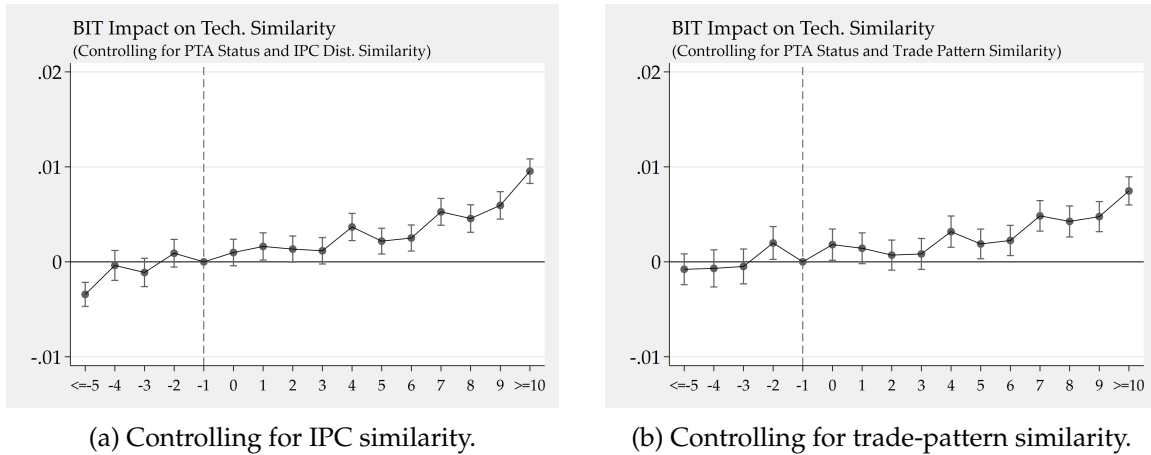


Figure 13: BIT effects on bilateral technology similarity.

Notes: Specification as in (3) with a BIT event indicator. Bars show 95% confidence intervals based on standard errors clustered at the country-pair level.



(a) Controlling for IPC similarity.

(b) Controlling for trade-pattern similarity.

Figure 14: BIT response with IPC and trade-pattern controls.

Notes: Panel (a) reports the BIT event-study response on  $S_{dot}$  with  $IPC4CosSim_{dot}$  as a contemporaneous control. Panel (b) reports the BIT response with the four directional trade-pattern cosine similarities and concurrent PTA status as controls. Specifications otherwise follow equation (3).

sector. BITs generate positive coefficients in regulation-intensive sectors, with the largest effects in chemicals (20), pharmaceuticals (21), rubber and plastics (22), fabricated metals (25), computers and electronics (26), and machinery repair and installation (33). These sectors substantially overlap with the regulation-intensive sectors identified for deep PTAs in Figure 11, consistent with investment-mediated integration operating through technological channels similar to those of deep-PTA-mediated integration.

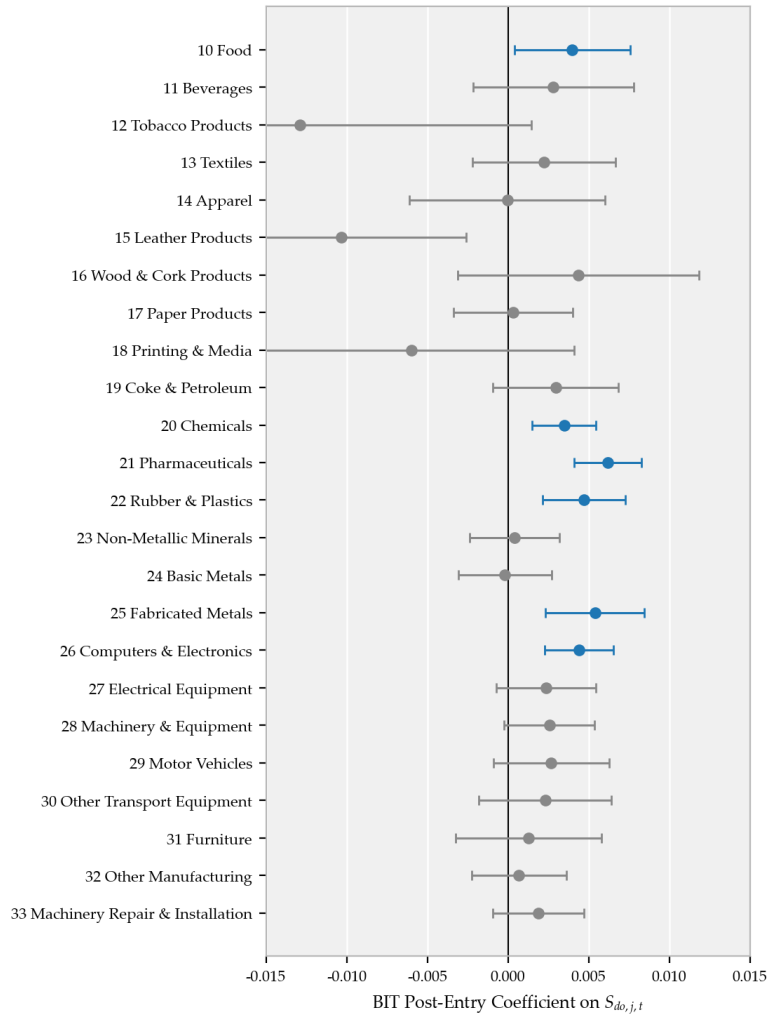


Figure 15: BIT effect by 2-digit ISIC sector.

*Notes:* Coefficients  $\hat{\beta}_k$  from the stacked sector-interaction BIT regression. Horizontal bars are 95% confidence intervals based on standard errors clustered at the country-pair-sector level. Blue markers denote coefficients positive and significant at the 5% level or better.

#### 4.4 WTO Accession

We turn next to WTO accession, which binds a country to a multilateral framework of market access, tariff bindings, and regulatory disciplines shared by all members. Both the direct integration that accession delivers and the domestic reforms that typically accompany it could pull countries' technologies toward those of other WTO members.

We estimate the same specification as before, with the event for a country pair  $(d, o)$  defined as the first year in which both countries are simultaneously members of the GATT or the WTO. Figure 16 reports the results: country pairs become more technologically

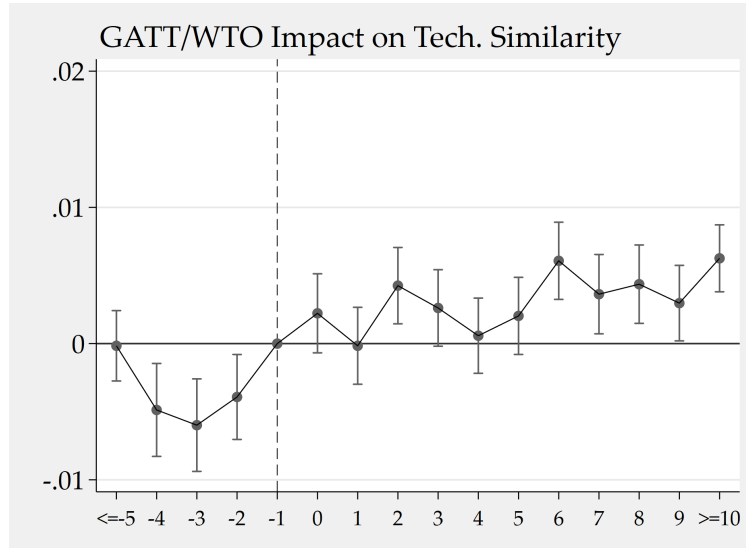


Figure 16: WTO/GATT accession effects on bilateral technology similarity.

Notes: Specification as in (3) with the GATT-or-WTO joint-membership event.

similar around and after joint accession, though the estimated coefficients are smaller than those for BITs and deep agreements.

A caveat in interpreting the results is that a couple of years prior to WTO/GATT accession, countries started converging in technology space. This could reflect as much the broad market-opening and regulatory reforms countries undertake during the accession process as the WTO rules themselves.

#### 4.5 Additional Evidence on the Mechanism

One reason deep agreements and BITs may move countries closer in technology space is that they reduce the need to design around separate technical systems. Technology standards harmonization provides a direct example of this channel: when partner countries participate in the same technology standards, or when standards-setting organizations whose members are more integrated issue more standards, firms face a common compatibility environment and can build technologies around the same interfaces. We therefore turn to standards data to provide evidence supportive of this mechanism.

We use data from the Searle Center Database on Technology Standards and Standard Setting Organizations (SSOs) (Baron and Spulber, 2018), which records SSO membership and bibliographic information on published standards over 1980–2014. For each SSO, we observe the member firms—the firms who use and can potentially shape the direction of the standards issued by the SSO. Using large language models combined with internet search, we find the origin country of these member firms. We then assign country weights

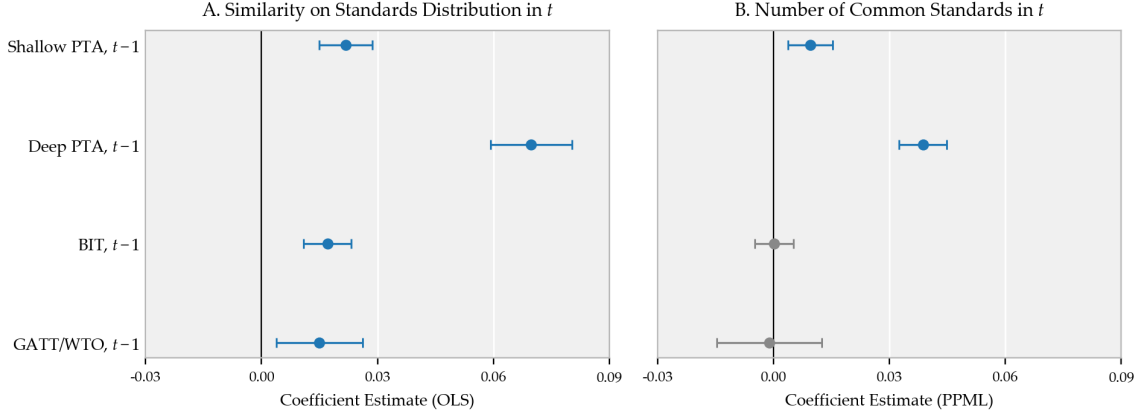


Figure 17: Agreement exposure and bilateral standards harmonization.

*Notes:* Panel A reports OLS coefficients from regressions of standards-distribution similarity on one lagged agreement indicator at a time. Panel B reports PPML coefficients from analogous regressions for the number of common standards. All specifications use one observation per unordered country pair-year and include country-pair, first-country-year, and second-country-year fixed effects. Bars are 95% confidence intervals based on standard errors clustered at the country-pair level. Blue markers denote coefficients whose 95% confidence interval lies above zero.

within an SSO according to the SSO's distribution of member firms across countries. We use this to construct a country's exposure to standards. For example, if 50% of an SSO's member firms are from China and 50% from the U.S., then both China and the U.S. are treated as 'owning' half of the standards issued by the SSO.

We aggregate this measure across standards issued by the SSOs in our dataset to construct measures of whether two countries are exposed to the same or similar set of standards. With these metrics at hand, we examine whether partners bound by multilateral agreements are exposed to more similar standards portfolios. Note that since our data contain a snapshot of SSO members, the variation we exploit is over time: as countries join multilateral institutions, we ask whether SSOs with firms from partner and non-partner countries experienced differential changes in their influence.

**Bilateral standards harmonization.** We first ask whether multilateral agreements are associated with convergence in the standards environment faced by partner countries. For each country-pair-year, we measure harmonization in two related ways: the cosine similarity between the partners' standard portfolios, constructed as described above, and the number of newly issued standards produced by SSOs in which both countries are represented. We regress each outcome on lagged agreement indicators, absorbing country-pair fixed effects and separate country-year fixed effects for each partner. Formally, following the notations in (3), for each agreement  $I \in \{\text{PTA shallow}, \text{PTA deep}, \text{BIT}, \text{WTO}\}$ , we run

$$Standards_{dot} = \beta \mathbb{I}[t-1 \geq T_{do}^I] + FE_{do} + FE_{dt} + FE_{ot} + \epsilon_{dot}. \quad (4)$$

Table 3: Agreement exposure and SSO expansion.

| Dep. Var.                                | Number of Newly Issued Standards |                     |                    |                   |
|------------------------------------------|----------------------------------|---------------------|--------------------|-------------------|
|                                          | (1)                              | (2)                 | (3)                | (4)               |
| Share of Pairs with Shallow PTA, $t - 1$ | -2.589<br>(1.881)                |                     |                    |                   |
| Share of Pairs with Deep PTA, $t - 1$    |                                  | 9.698***<br>(1.147) |                    |                   |
| Share of Pairs with BIT, $t - 1$         |                                  |                     | 10.039*<br>(5.567) |                   |
| Share of Pairs with GATT/WTO, $t - 1$    |                                  |                     |                    | -3.455<br>(2.551) |
| SSO FE                                   | ✓                                | ✓                   | ✓                  | ✓                 |
| Year FE                                  | ✓                                | ✓                   | ✓                  | ✓                 |
| Observations                             | 647                              | 647                 | 647                | 647               |
| Pseudo $R^2$                             | 0.943                            | 0.952               | 0.949              | 0.944             |

Notes: Each column reports a separate PPML regression at the SSO-year level. The dependent variable is the number of standards newly issued by an SSO in a year. The agreement variable is the lagged weighted share of country pairs in the SSO that are covered by the agreement, where country-pair weights are the products of the two countries' shares in the SSO's membership. All regressions include SSO and year fixed effects. Standard errors clustered at the SSO level are reported in parentheses. \*,  $p < 0.1$ ; \*\*,  $p < 0.05$ ; \*\*\*,  $p < 0.01$ .

where  $Standards_{dot}$  is either the cosine similarity of the two countries' standards distributions, estimated with OLS, or the count of common standards, estimated with PPML. The agreement indicator is lagged by one year to mitigate concerns about reverse causality. If multilateral agreements enhance the standard harmonization among agreement countries, we should expect the coefficient to be positive, resulting from additional standard issuance by SSOs with agreement countries.

Figure 17 reports the results. Panel A shows that agreement exposure is associated with greater similarity in partners' standards distributions, with deep PTAs clearly standing out. The deep-PTA coefficient is 0.070 on the cosine-similarity scale, compared with 0.022 for shallow PTAs, 0.017 for BITs, and 0.015 for joint GATT/WTO membership. Panel B shows the same ranking at the count margin: the PPML coefficient for deep PTAs is 0.039, implying about 4 percent more common standards, whereas the shallow-PTA coefficient is about 1 percent and the BIT and GATT/WTO coefficients are essentially zero. The standards response is therefore not a generic consequence of international agreements; it is concentrated where agreements create richer regulatory and technical commitments, consistent with standards harmonization as one mechanism behind the deep-PTA technology effect.

**SSO expansion.** We then turn to the standards organization level and ask whether SSOs whose member countries are more institutionally integrated gain prominence, as

measured by the number of standards they issue. For SSO  $g$ , let  $w_{gi}$  be the static membership weight of country  $i$  in that SSO, defined as the share of member firms in the SSO that are from country  $i$ . We then construct a lagged SSO-year exposure measure for each agreement type  $I \in \{\text{PTA shallow}, \text{PTA deep}, \text{BIT}, \text{WTO}\}$  as:

$$X_{g,t-1}^I = \frac{\sum_{i < j} w_{gi} w_{gj} \mathbb{I}[t-1 \geq T_{ij}^I]}{\sum_{i < j} w_{gi} w_{gj}}. \quad (5)$$

This measure can be viewed as the weighted share of represented country pairs in SSO  $g$  that are covered by agreement  $I$  in year  $t-1$ . Let  $StdNum_{gt}$  be the number of standards issued by SSO  $g$  in year  $t$ . We run

$$StdNum_{gt} = \beta X_{g,t-1}^I + FE_g + FE_t + \epsilon_{gt}, \quad (6)$$

estimated by PPML with SSO and year fixed effects.

Table 3 reports the results. Deep-PTA exposure continues to hold a robust positive association: a one-percentage-point increase in the weighted deep-PTA share corresponds to approximately  $\exp(0.097) - 1 \simeq 10\%$  more standards issued in a year. BIT exposure is similar in magnitude but less precise, while shallow-PTA and GATT/WTO exposure are not positive. The SSO-level evidence therefore mirrors the bilateral evidence: the standards system expands most strongly when represented countries are tied by deeper regulatory agreements.

## 4.6 Summary

Our findings suggest that deep PTAs generate the largest and most immediate increase in technology integration. The effect operates on both the across-field and within-field margins and survives controls for the similarity of partners' import and export compositions, suggesting this is more than a story of countries aligning in global production networks and hence becoming more similar in their economic structure. We also find BITs to have qualitatively similar but more gradual effects. Both BITs and deep PTAs raise bilateral technology similarity in a shared set of regulation and compatibility intensive sectors—chemicals, pharmaceuticals, computers and electronics, and machinery repair and installation—which lends further support to the regulatory-cooperation and investment-mediated channels we discussed above. After joint WTO/GATT accession, country pairs become more technologically similar, but the effects are smaller.

An explanation consistent with these findings is as follows. By opening common markets and stabilizing the policy environment, multilateral agreements raise the return to innovation that is applicable across partner countries. They also give firms a stake in the

standard-setting that keeps those technologies interoperable. Deep agreements reinforce this channel. By harmonizing standards and aligning regulations, they make a common technical environment easier to build. In support of this explanation, the standard-setting evidence in Section 4.5 shows that SSOs whose members are tied by deep agreements issue more standards. The effect is strongest for deep PTAs and BITs and weak for shallow PTAs and the WTO. This is the same ranking we find in the technology-similarity results.

## 5 Counterfactual Analysis

We translate the event-study estimates of the impacts of multilateral agreements on technology interdependence into counterfactual bilateral technology realignments. The analysis proceeds in two steps. The first is backward-looking and decomposes the observed technology alignment over 1980–2020 into the contribution of PTAs, BITs, and the GATT/WTO, both globally and for the two case studies of Section 3. The second is forward-looking and centered on the US–China power competition: we simulate technology-space paths under two alternative architectures of preferential agreements involving the United States and China.

To implement the first set of simulations, we reverse the long-run effect of every treaty change the dyad underwent strictly after 1980. We use the long-run coefficients from Section 4:  $\beta_{\text{shallow}}^{\text{PTA}} = 0.0025$ ,  $\beta_{\text{deep}}^{\text{PTA}} = 0.0104$ ,  $\beta^{\text{BIT}} = 0.0052$ , and  $\beta^{\text{GATT/WTO}} = 0.0047$ . Each is the unweighted mean of the post-event point estimates at horizons  $+6$  to  $\geq +10$ . The two PTA coefficients are mutually exclusive at the dyad level: a dyad in a deep PTA is adjusted by  $\beta_{\text{deep}}^{\text{PTA}}$  and a dyad in a shallow PTA only by  $\beta_{\text{shallow}}^{\text{PTA}}$ . Pairs whose treaty status was unchanged after 1980 receive no adjustment, so realized and counterfactual paths coincide at 1980 by construction. A dyad that was upgraded from a shallow PTA to a deep PTA after 1980 is adjusted by  $\beta_{\text{deep}}^{\text{PTA}} - \beta_{\text{shallow}}^{\text{PTA}}$ , the upgrade increment.

We implement the second set of simulations analogously, by adding or subtracting these coefficients to pairs of countries according to assumed dyad-level treaty status as of 2021.

### 5.1 Accounting for Past Outcomes

**Global trend.** We compute average bilateral technology similarity across all directed country pairs in non-overlapping five-year bins from 1981–1985 through 2016–2020; binning smooths short-run measurement noise without averaging away the long-run trend. The realized average rises by 0.048 in the actual data, from 0.560 in 1981–1985 to 0.608 in 2016–2020.



Table 4: Decomposition of the change in average bilateral technology similarity, 1981–1985 to 2016–2020.

|                                                         | Total Change | Share  |
|---------------------------------------------------------|--------------|--------|
| <i>Panel A. All country pairs</i>                       |              |        |
| Total (benchmark)                                       | 0.048        | 100.0% |
| Accounted for by PTA                                    |              | 1.8%   |
| Accounted for by BIT                                    |              | 1.1%   |
| Accounted for by GATT/WTO                               |              | 4.1%   |
| <i>Panel B. Within pairs that ever had the relation</i> |              |        |
| Accounted for by PTA                                    |              | 2.7%   |
| Accounted for by BIT                                    |              | 2.8%   |
| Accounted for by GATT/WTO                               |              | 4.2%   |

Notes: Panel A reports each relation’s share of the global rise across all country pairs. Panel B restricts to the subset of pairs that ever held the relation in question and reports each relation’s share of the rise within its own subset; the within-subset rise differs across rows and is itself larger than the global rise.

Panel A of Table 4 reports the decomposition of this rise: shutting down all post-1980 PTA, BIT, and GATT/WTO events together accounts for 7.0%; individually, the GATT/WTO floor of joint membership accounts for 4.1%, PTAs (shallow and deep combined) for 1.8%, and BITs for 1.1%. Restricting to country pairs that ever held each relation (Panel B), the within-subset rise is itself larger and the relation accounts for a larger fraction: 2.7%, 2.8%, and 4.2% of that within-subset rise respectively.

**Eastern Europe and the post-Soviet space.** For the Eastern European cohort of Figure 4, we re-compute the projection onto the Western-Europe–Russia line in the 2016–2020 bin after switching off all post-1980 PTA, BIT, and GATT/WTO events for every country pair in the dataset. Figure 18 contrasts the realized positions (black dots) with the counterfactual ones (stars): blue countries shift toward Western Europe under the counterfactual, while orange countries shift toward Russia. Bulgaria, Hungary, and Poland—each currently within the neighborhood of the Western European pole—move *away* from it in the counterfactual, by 3–6% of their remaining distance to Russia. Without the EU-accession-and-Europe-Agreement architecture, these economies would still sit on the Western half of the line, but their technological proximity to Western Europe would be visibly weaker. Belarus and Moldova move in the opposite direction, drifting toward Western Europe by 5–6% of their distance to Russia, reflecting their post-1980 treaty integration with Russia—the Belarus–Russia Union State for Belarus, and joint WTO accession for Moldova—that anchors them on the Russian side under the realized series. The counterfactual exercise thus reads east-west alignment in the post-Soviet space as a two-sided treaty story amid the Western pull.

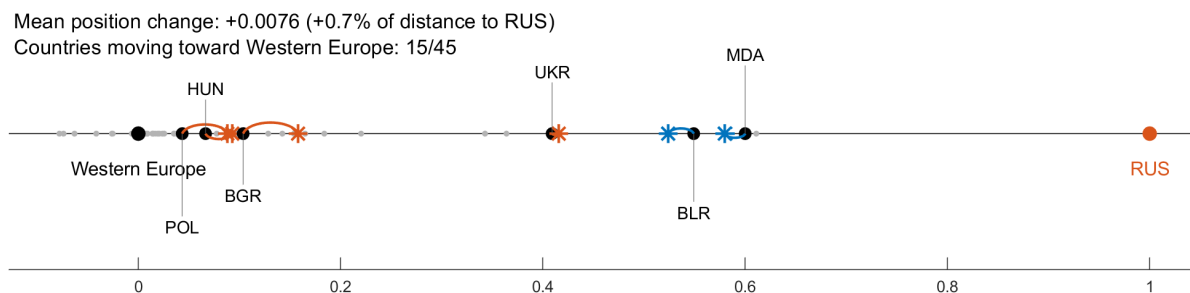


Figure 18: Counterfactual positions on the Western-Europe–Russia axis absent post-1980 events.

*Notes:* Black dots are the realized 2016–2020 positions on the Western-Europe–Russia line; stars are the counterfactual positions after switching off all post-1980 PTA, BIT, and GATT/WTO events for every country pair in the dataset. Blue countries move toward Western Europe under the counterfactual, while orange countries move toward Russia.

## 5.2 Evaluating Future Scenarios

We now turn to two prospective architectures of global geopolitical arrangement whose technology-space implications bear directly on the ongoing US–China realignment, evaluated against the realized 2019–2021 distribution of countries on the US–China line. The two cases differ in which power leads the realignment and which loses ground.

**Case 1: China-led realignment.** China accedes to the CPTPP<sup>10</sup> and the RCEP<sup>11</sup>, gaining a deep PTA with each member with which it does not currently have one; in parallel, the United States exits all of its existing PTAs and the GATT/WTO. Figure 19, upper panel, plots the resulting position shifts on the US–China axis (red stars). The average highlighted economy moves +0.025 toward China, equivalent to +3.3% of its remaining distance to the Chinese pole. The shift is driven mainly by the United States’ withdrawal from the GATT/WTO floor it currently anchors: every country with WTO ties to the United States loses that anchor and drifts toward China. Germany, France, the United Kingdom, and Russia all move by the same percentage (+2.6%, the WTO-only channel); Japan, addition-

<sup>10</sup>The Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP), signed 8 March 2018 and entered into force 30 December 2018, succeeded the Trans-Pacific Partnership (TPP) after the United States withdrew on 23 January 2017. Signatory members: Australia, Brunei, Canada, Chile, Japan, Malaysia, Mexico, New Zealand, Peru, Singapore, and Vietnam. China formally applied to join on 16 September 2021. We use the eleven CPTPP signatories as the hypothetical accession set for both China (Case 1) and the United States (Case 2).

<sup>11</sup>The Regional Comprehensive Economic Partnership (RCEP), signed 15 November 2020, entered into force on 1 January 2022 for ten of its fifteen signatories. Members: Australia, Brunei, Cambodia, China, Indonesia, Japan, South Korea, Laos, Malaysia, Myanmar, New Zealand, the Philippines, Singapore, Thailand, and Vietnam. China deposited its instrument of ratification on 15 April 2021. We model the hypothetical as further deep agreements between China and the other RCEP members. That is, China joins the agreement and forms deep PTAs with all members.

ally a beneficiary of CHN's deep accession to the CPTPP, moves +6.0%; Korea, the closest of the highlighted economies to the Chinese pole and a beneficiary of both the WTO drop and CHN's deep accession through the RCEP, moves the most in proportional terms (+9.9%). Russia, already close to the Chinese pole, shifts less in absolute terms because the Western anchoring it loses through the US's WTO exit is small to begin with.

**Case 2: US-led realignment.** The United States accedes to the CPTPP and to a revived TTIP<sup>12</sup>, gaining a deep PTA with each CPTPP signatory and each EU member state; China is excluded from all PTAs and the GATT/WTO. Figure 19, lower panel, shows the resulting position shifts (blue stars). The average highlighted economy moves  $-0.026$ , or  $-3.7\%$  of its distance to the Chinese pole. Russia, with joint WTO membership with China and substantial pre-existing technology proximity, moves the most in proportional terms ( $-12.9\%$ ). Japan moves  $-6.7\%$ , reflecting the new USA–Japan deep PTA via the CPTPP combined with the loss of CHN's WTO membership. Korea moves  $-5.6\%$  through the reversal of both the CHN–KOR FTA and its joint WTO membership with China. The European economies—Germany, France, and the United Kingdom—move by 4.9–5.4%, almost entirely through the new TTIP deep PTA with the United States.

Taken together, the two scenarios illustrate the geographic asymmetry of the US–China realignment as observed through technology space. The two architectures move the average partner economy by similar proportional magnitudes, but the geopolitical incidence differs and crucially depends on existing ties. A China-led realignment, in which Beijing extends deep agreements to its Asian neighbors via the RCEP and the CPTPP while Washington withdraws from the multilateral system, pulls Asian economies the most while Western European economies move only by the small WTO-only channel. A US-led realignment, in which Washington restores deep transatlantic and trans-Pacific integration via the CPTPP and the TTIP while China is excluded from the multilateral system, instead reaches Western Europe meaningfully through the TTIP: Germany, France, and the United Kingdom move significantly away from China, in addition to large shifts for Russia, Japan, and Korea via their existing China ties.

A caveat in interpreting this result is that, while it shows different alignment scenarios can lead to different technology positions, it is silent on the true economic impact of these positions—on trade and ultimately welfare. It is also silent on general equilibrium effects: when one agreement is weakened and some countries shift their position in the technology space as a result, countries that are not directly affected by the agreement would also react.

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<sup>12</sup>The Transatlantic Trade and Investment Partnership (TTIP) was negotiated between the United States and the European Union from July 2013, stalled after the change of US administration in early 2017, and was formally declared obsolete by the EU Council in April 2019. We model the hypothetical “restored TTIP” as a deep agreement covering the United States and the 28 EU member states at the time of negotiation (including the United Kingdom at the time).

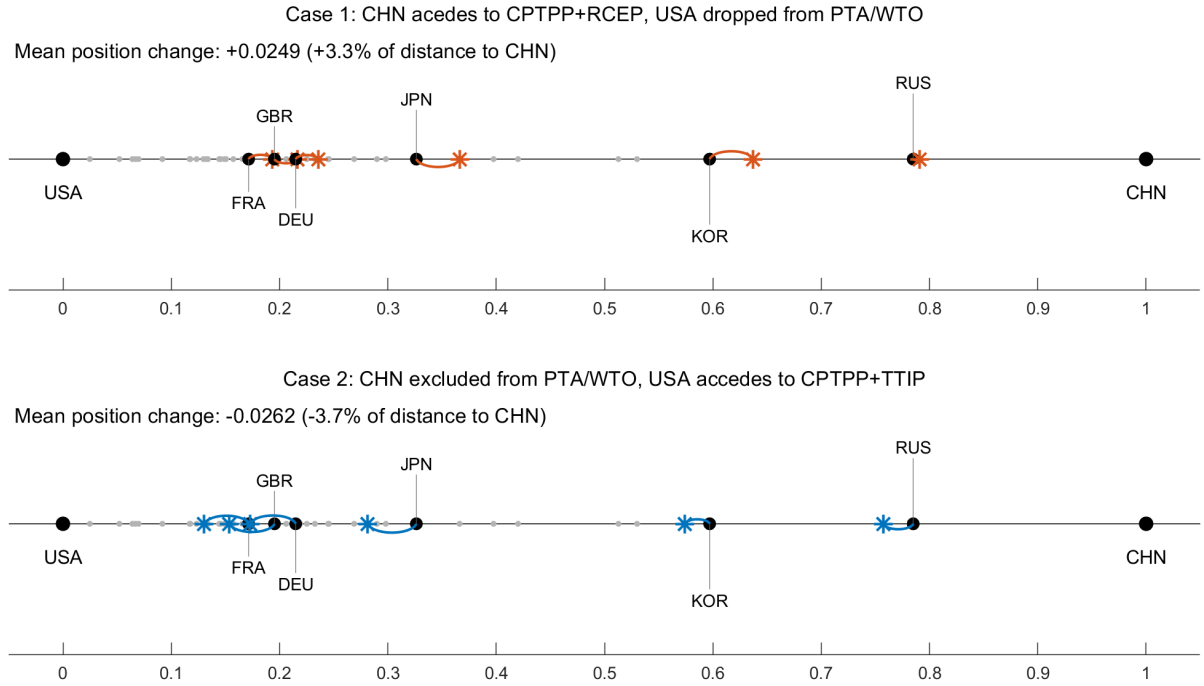


Figure 19: Counterfactual positions on the US–China axis under two realignment scenarios, 2019–2021.

*Notes:* Black dots are the realized 2019–2021 positions and stars are the counterfactual ones. Upper panel: China-led realignment, with China acceding to the CPTPP and the RCEP and the United States exiting all PTAs and the GATT/WTO. Lower panel: US-led realignment, with the United States acceding to the CPTPP and the restored TTIP and China excluded from all PTAs and the GATT/WTO.

Thus, the findings are best interpreted as the partial-equilibrium response, holding the unaffected countries fixed.

## 6 Conclusion

This paper examines the historical role of multilateral agreements in shaping countries' positions in technology space. Using a bilateral technology similarity measure built from patent semantics and covering 1980–2020, we document that the integration era built technological compatibility across countries based on country-specific economic and geopolitical alignment. Two historical episodes illustrate this directly: after the collapse of the Soviet Union, Hungary, Poland, and Bulgaria moved from intermediate positions in the early 1990s toward the Western European pole alongside their Europe Agreements and EU accession, whereas Belarus and Moldova remained relatively closer to Russia; after China's

1992 Southern Tour, China's technologies drifted toward the United States through the 2000s and then drifted away after the mid-2010s when tensions between the two countries started to rise, while traditional U.S. allies such as Germany and Japan remained close to the United States throughout.

The event-study analysis shows that deep PTAs generate the largest and most persistent increase in bilateral technology similarity, operating on both across-field and within-field margins and surviving controls for the similarity of partners' import and export compositions. BITs deliver qualitatively similar but more gradual effects. Both BITs and deep PTAs raise bilateral technology similarity most in industries where technical standards are most relevant—chemicals, pharmaceuticals, computers and electronics, and machinery repair and installation—which lends further support to the regulatory-cooperation and investment-mediated channels. After joint WTO/GATT accession, country pairs likewise become more technologically similar, though the effects are smaller and could be partly related to the broad market-opening and regulatory reforms countries undertake during accession as much as the WTO rules themselves.

Building on these estimates, we conduct counterfactual simulations. The integration-era agreements account for 7.0% of the rise in average bilateral technology similarity between 1981–1985 and 2016–2020, with the GATT/WTO floor of joint membership accounting for 4.1%, PTAs for 1.8%, and BITs for 1.1%. Looking forward, a China-led realignment, in which China deepens its CPTPP and RCEP linkages while the United States exits the multilateral system, would push the average highlighted economy by +3.3% of its remaining distance toward the Chinese pole, with the largest pulls on its Asian neighbors. A mirror US-led realignment, in which Washington restores deep integration through the CPTPP and a revived TTIP while China is excluded from the multilateral system, moves the same economies –3.7% in the opposite direction, reaching Western Europe meaningfully through the TTIP.

The results speak to the current international environment, in which the multilateral agreements that shaped the integration era are being challenged by dominant economies. Our empirical and counterfactual evidence suggests that these agreements were quantitatively important in shaping countries' positions in technology space. To the extent that interoperable technologies support the integration of global value chains, facilitate the adoption of new technologies, and encourage innovation by expanding markets, the fragmentation of the multilateral order could affect welfare in ways that extend beyond its direct effect on trade and FDI.

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## A Construction of the Technology Interdependence Measure

This subsection collects the details of the measure summarized in Section 2, including the construction from patent texts and the data-processing choices that underlie the coverage, the embedding computation, and the technology-cluster exercise of Section 2.2.

We measure technology interdependence between a pair of units (firms, sectors, or countries) by the cosine similarity of their average patent-text embeddings. Embeddings are produced by a sentence-encoder language model applied to the text of each patent’s abstract, and averaging is done over the patents belonging to each unit. This similarity measure, as we show in Section 2.2 using the example of vehicle powertrains and their upstream components, tracks technological compatibility along a production chain and discriminates between alternative technology paths inside the same broad technology class. In the regression analysis we rely on over-time variation in the measure, always controlling for country-pair, technology-class-pair, or sector-pair fixed effects, so that any semantic similarity attributable to time-invariant components of the bilateral relation (shared patent-writing conventions, geographic and linguistic proximity) is differenced out.

We build the measure from the 2023 Autumn Edition of *PATSTAT Global*, maintained by the European Patent Office, which covers close to the universe of world patents.<sup>13</sup> For consistency in language processing, we keep patents whose abstracts are in English, covering roughly 90% of all abstracts in PATSTAT. To avoid double-counting the same invention filed across multiple offices, we retain one patent per patent family. The final sample contains roughly 45 million patent abstracts.

Each abstract is converted into a numerical vector using a sentence-encoder language model, a class of large language models that map a paragraph into a fixed-length real vector whose geometry reflects semantic meaning: two paragraphs with similar content map to vectors that point in similar directions under cosine similarity. The Sentence-BERT recipe of Reimers and Gurevych (2019) introduced siamese fine-tuning of transformer encoders to produce semantically meaningful sentence embeddings, and the resulting class of models is now standard for semantic retrieval (Karpukhin et al., 2020), document clustering (Grootendorst, 2022), and textual-similarity assessment (Gao et al., 2021).<sup>14</sup> Specifically, we use a performant sentence-encoder model, *gte-base-en-v1.5*, developed by Alibaba, which produces a 768-dimensional embedding for each abstract.<sup>15</sup> We denote by

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<sup>13</sup>PATSTAT aggregates records from about 90 patent-issuing authorities, including major national offices such as the United States Patent and Trademark Office (USPTO) and the China National Intellectual Property Administration (CNIPA), regional offices such as the European Patent Office (EPO), and international publications under the Patent Cooperation Treaty (PCT).

<sup>14</sup>The models are fine-tuned on natural-language-inference and paraphrase data so that the pooled output of a transformer encoder, when compared across two texts, approximates human judgments of semantic similarity. The output vector is commonly referred to as the *embedding* of the text.

<sup>15</sup>The model performs well on the Massive Text Embedding Benchmark (Muennighoff et al., 2023): as of

$e_p \in \mathbb{R}^{768}$  the embedding of patent  $p$ ; viewed as a point in the 768-dimensional embedding space,  $e_p$  summarizes the combination of technical components, scientific principles, and engineering problems that define the invention.

The empirical analysis uses country, country-IPC4, and country-sector pairs as the units of analysis. We aggregate IPC4s into two-digit sectors using a concordance from IPC to the standard industrial classification. We crosswalk patents to countries by the nationality of inventors listed on each patent.<sup>16</sup> Letting  $u$  denote a generic unit (a country, or a country-sector) and  $P_{u,t}$  the set of patents associated with  $u$  in period  $t$ , we represent the technology of  $u$  at  $t$  by the weighted average embedding

$$\bar{e}_{u,t} = \frac{\sum_{p \in P_{u,t}} w_{p,u} e_p}{\sum_{p \in P_{u,t}} w_{p,u}}, \quad (7)$$

where  $w_{p,u}$  is the product of the inventor-country share and the IPC-to-sector share of patent  $p$  accruing to  $u$ . The bilateral technology interdependence between two units  $u$  and  $u'$  is the cosine similarity of their average embeddings,

$$\text{Similarity}_{uu',t} = \frac{\bar{e}_{u,t} \cdot \bar{e}_{u',t}}{\|\bar{e}_{u,t}\| \|\bar{e}_{u',t}\|} \in [-1, 1]. \quad (8)$$

A value close to one indicates that the two units draw on the same set of technological components, a value close to zero that they are technologically unrelated, and a value close to negative one that they point in opposite technological directions. An alternative aggregation would be to compute pairwise cosine similarities across individual patents and then average. At our scale this pairwise approach is computationally prohibitive; on subsamples where both are feasible, bootstrap resampling at the patent level shows that the centroid-based measure is close to the pairwise-averaged alternative, while being orders of magnitude cheaper to compute.

**Patent Data Processing.** We start from PATSTAT Global (2023 Autumn Edition). We keep patents whose abstracts are in English and retain one patent per DOCDB family, taking as the family representative the earliest filing. A patent can belong to multiple IPC subclasses; we record the full list and attach to each (patent, IPC4) record an IPC4 weight equal to the inverse of the number of IPC4 codes listed on the patent. Inventor country is taken from the country codes of inventors listed on the family representative; when more than one

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August 2024 it ranks 36 out of 438 models on the public leaderboard, while the default model of Sentence-BERT (all-mpnet-base-v2) ranks 282nd and OpenAI's text-embedding-3-large ranks 65th.

<sup>16</sup>A patent can list several inventor countries and several IPC subclasses. When constructing the average embedding of a country-sector, we split each patent's contribution across countries in proportion to the number of inventors located in each, and across IPC subclasses in proportion to the crosswalk weights from the IPC to the sector.

country is listed, we assign fractional weights proportional to the number of inventors in each country. For patents with no inventor listed, we assign the patent to the country of the patent office of first filing. All aggregations to the country-sector or country-product level use these two weights.

**Embedding Computation.** Embeddings are computed with `gte-base-en-v1.5` (Alibaba), accessed through the `sentence-transformers` Python library. The model supports a maximum context length of 8,192 tokens. We run the model in single precision (`float32`) on a single NVIDIA GeForce RTX 4090 GPU, at a throughput of roughly 25 million abstracts per day; processing all 45 million abstracts takes about two days. Embeddings are stored in yearly Parquet shards.

**Technology-Cluster Construction.** For each IPC4 with at least 20 patents, we pool all patents across years 1960–2023 and fit  $K$ -means on their embeddings. The number of clusters  $K$  is determined by the size  $n$  of the class according to

$$K = \max\{5, \min\{30, \lfloor \sqrt{n/20} \rfloor\}\},$$

so that  $K$  is bounded between 5 and 30 and, between these caps, grows with the square root of the class size. This rule gives larger IPC4s a finer partition while keeping cluster sizes comparable across classes.

**TF-IDF Cluster Keywords.** To label each cluster, we apply a term-frequency-inverse-document-frequency (TF-IDF) vectorizer to the concatenated abstracts of each cluster, treating sibling clusters within the same IPC4 as the document collection. Formally, for a given IPC4 with  $C$  clusters, we concatenate the patent abstracts of cluster  $c$  into a single document  $d_c$ . For each term  $w$  (a unigram, bigram, or trigram after removing English stop words), the term frequency is the number of occurrences of  $w$  in  $d_c$ , denoted  $\text{TF}(w, c)$ , and the inverse document frequency is

$$\text{IDF}(w) = \log \frac{1 + C}{1 + \#\{c : w \in d_c\}} + 1,$$

where  $\#\{c : w \in d_c\}$  is the number of documents containing  $w$ . The TF-IDF score of  $w$  in cluster  $c$  is  $\text{TF-IDF}(w, c) = \text{TF}(w, c) \cdot \text{IDF}(w)$ . A word that occurs often in cluster  $c$  but rarely in the other clusters of the IPC4 receives a high score; a word that is common across all clusters (e.g., “vehicle” in B60K) receives a low score because its IDF is small. We further drop any term appearing in more than 80% of the clusters in the IPC4 to remove common

terms that are not related to technology information. We then rank words in each cluster by their TF-IDF score and report the top-ranked words as the cluster's keywords.